

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent

12411893

Kind Code

B2

Date of Patent

September 09, 2025

Inventor(s)

Parasnis; Abhay et al.

Proactively generated content and personalized feeds available for audiences

Abstract

Methods, systems, and computer programs are presented for generating personalized content. One method includes an operation for identifying audience parameter values that indicate which users are members of an audience. The method further includes operations for determining attribute values for attributes used to generate items based on the audience parameter values, and generating items for the audience based on the attribute values. Generating each item comprises creating a prompt based on a type of the item, the attribute values, and the audience parameter values; selecting a generative artificial intelligence (GAI) tool to generate the item based on the type of item; and providing the created prompt to the selected GAI tool. The method further includes causing presentation of the generated items in a user interface (UI), receiving on the UI a selection of one of the items; and transmitting the selected item to one or more members of the audience.

Inventors: Parasnis; Abhay (Palo Alto, CA), Shah; Saachi (Palo Alto, CA), Moreira; Jonathan (Palo Alto, CA), Bendapudi; Perraju (Palo Alto, CA), Pola; Hema Vishnu (Palo Alto, CA), Sood; Vishal (Palo Alto, CA), Dave; Nipun (Palo Alto, CA), Pimpley; Anish (Palo Alto, CA)

Applicant: Typeface Inc. (Palo Alto, CA)

Family ID: 96022543

Assignee: Typeface Inc. (Palo Alto, CA)

Appl. No.: 18/783424

Filed: July 25, 2024

Prior Publication Data

Document Identifier

US 20250200114 A1

Publication Date

Jun. 19, 2025

Related U.S. Application Data

continuation parent-doc US 18203537 20230530 US 12045735 child-doc US PCT/US2024/012269
continuation parent-doc US 18203524 20230530 US 11809688 child-doc US PCT/US2024/012269
continuation parent-doc US 18203534 20230530 US 11922541 child-doc US PCT/US2024/012269
continuation parent-doc US 18203530 20230530 US 11928319 child-doc US PCT/US2024/012269
continuation-in-part parent-doc WO PCT/US2024/012269 20240119 PENDING child-doc US 18783424
us-provisional-application US 63644385 20240508
us-provisional-application US 63637258 20240422
us-provisional-application US 63637277 20240422
us-provisional-application US 63637254 20240422
us-provisional-application US 63637275 20240422
us-provisional-application US 63637266 20240422
us-provisional-application US 63611006 20231215
us-provisional-application US 63444162 20230208

Publication Classification

Int. Cl.: G06F16/00 (20190101); G06F3/04842 (20220101); G06F16/9035 (20190101);
G06Q30/0242 (20230101); G06Q30/0251 (20230101)

U.S. Cl.:

CPC G06F16/9035 (20190101); G06F3/04842 (20130101); G06Q30/0244 (20130101);
G06Q30/0269 (20130101);

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8316381	12/2011	Fearn	714/47.1	G06F 11/3495
8655889	12/2013	Hua et al.	N/A	N/A
8717367	12/2013	Clifton et al.	N/A	N/A
9411830	12/2015	Mei et al.	N/A	N/A
9501499	12/2015	Sanio	N/A	G06F 16/951
10032188	12/2017	Jakobi	N/A	G06Q 30/0253
10268669	12/2018	Allen	N/A	N/A
10719566	12/2019	Zhang	N/A	G06F 16/907
10839418	12/2019	Aragonda	N/A	G06Q 30/0249
10991012	12/2020	Balakrishnan et al.	N/A	N/A

11392659	12/2021	Jain et al.	N/A	N/A
11417085	12/2021	Saraee	N/A	G06V 10/761
11516158	12/2021	Luzhnica et al.	N/A	N/A
11550884	12/2022	Kumar et al.	N/A	N/A
11596867	12/2022	Snodgrass	N/A	A63F 13/56
11809688	12/2022	Parasnis	N/A	G06F 3/04845
11922541	12/2023	Parasnis et al.	N/A	N/A
11928319	12/2023	Parasnis	N/A	G06T 5/77
11995803	12/2023	Karpman et al.	N/A	N/A
12045735	12/2023	Parasnis et al.	N/A	N/A
2007/0162845	12/2006	Cave	715/764	G06F 40/117
2008/0215979	12/2007	Clifton	707/E17.009	G06F 16/40
2012/0054004	12/2011	Priyadarshan	705/14.4	G06Q 30/0241
2012/0110432	12/2011	Mei et al.	N/A	N/A
2014/0096026	12/2013	Grosz	N/A	N/A
2015/0112980	12/2014	Sanio	707/728	G06F 16/532
2015/0248429	12/2014	Pregueiro	715/202	G06T 11/60
2015/0271540	12/2014	Melby	725/34	H04N 21/25891
2016/0140622	12/2015	Wang	705/14.66	G06Q 50/01
2016/0140623	12/2015	Gupta	705/14.66	G06Q 30/0204
2016/0225014	12/2015	Mitra	N/A	G06Q 30/0243
2016/0314502	12/2015	Lindsay et al.	N/A	N/A
2017/0180443	12/2016	Raichelgauz et al.	N/A	N/A
2017/0206190	12/2016	Satterfield et al.	N/A	N/A
2017/0287013	12/2016	Iannaccone	N/A	G06Q 30/0277
2018/0018707	12/2017	Berry, Jr.	N/A	G06Q 30/0257
2018/0040019	12/2017	Gavlovski	N/A	G06F 16/24578
2018/0082329	12/2017	Johnson	N/A	G06Q 30/0275
2018/0268490	12/2017	Sodomka	N/A	G06Q 40/12
2018/0374253	12/2017	Yu et al.	N/A	N/A
2019/0034973	12/2018	Yao	N/A	G06Q 50/01
2019/0114664	12/2018	Goldman	N/A	G06Q 30/0277
2019/0325088	12/2018	Dubey et al.	N/A	N/A
2019/0377460	12/2018	Butcher et al.	N/A	N/A
2020/0125609	12/2019	Cetintas	N/A	G06F 16/3331
2020/0273062	12/2019	Probell	N/A	N/A
2020/0279008	12/2019	Jain et al.	N/A	N/A
2020/0285703	12/2019	Ramachandra	N/A	N/A

2020/0410026	12/2019	Reuter	N/A	G06F 16/9536
2020/0410552	12/2019	Stohlman	N/A	G06N 5/025
2021/0026522	12/2020	Bowen	N/A	G06F 30/00
2021/0042796	12/2020	Khoury et al.	N/A	N/A
2021/0118012	12/2020	Chung	N/A	N/A
2021/0146254	12/2020	Snodgrass	N/A	G06N 3/08
2021/0201360	12/2020	Randazzo	N/A	G06F 16/906
2021/0224858	12/2020	Khoury et al.	N/A	N/A
2021/0382956	12/2020	Reuter	N/A	G06F 16/9536
2022/0005070	12/2021	Raviv	N/A	G06Q 30/0206
2022/0005235	12/2021	Gou et al.	N/A	N/A
2022/0108417	12/2021	Liu et al.	N/A	N/A
2022/0148059	12/2021	Faricy et al.	N/A	N/A
2022/0179899	12/2021	Dong	N/A	N/A
2022/0198779	12/2021	Saraee et al.	N/A	N/A
2022/0244818	12/2021	Strecher et al.	N/A	N/A
2022/0301118	12/2021	Frey et al.	N/A	N/A
2022/0357914	12/2021	Krishnamurthy et al.	N/A	N/A
2022/0366435	12/2021	Boatner et al.	N/A	N/A
2022/0398379	12/2021	Hou et al.	N/A	N/A
2022/0414320	12/2021	Dolan et al.	N/A	N/A
2023/0005108	12/2022	Gopalkrishna et al.	N/A	N/A
2023/0013199	12/2022	Goldman	N/A	G06Q 30/0246
2023/0021797	12/2022	Gupta et al.	N/A	N/A
2023/0104661	12/2022	Wu et al.	N/A	N/A
2023/0119466	12/2022	Sjölander et al.	N/A	N/A
2023/0143430	12/2022	Vanreusel et al.	N/A	N/A
2023/0152598	12/2022	Brebner et al.	N/A	N/A
2023/0186535	12/2022	Malur Srinivasan et al.	N/A	N/A
2023/0214583	12/2022	Sawyer et al.	N/A	N/A
2023/0230198	12/2022	Zhang et al.	N/A	N/A
2023/0237980	12/2022	Pajjuri et al.	N/A	N/A
2024/0020863	12/2023	Zhu et al.	N/A	N/A
2024/0086648	12/2023	Han	N/A	G06N 3/08
2024/0127292	12/2023	Estes	N/A	G06F 16/9535
2024/0129601	12/2023	Brdiczka	N/A	H04N 21/854
2024/0249318	12/2023	Spiegel	N/A	H04L 51/02
2024/0256337	12/2023	Verma et al.	N/A	N/A
2024/0265274	12/2023	Parasnis et al.	N/A	N/A
2024/0273308	12/2023	Cai et al.	N/A	N/A
2024/0289396	12/2023	Chrysanthou	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2004334337	12/2003	JP	N/A
2013075316	12/2012	WO	N/A
2023069561	12/2022	WO	N/A
2024167650	12/2023	WO	N/A
2024167651	12/2023	WO	N/A

OTHER PUBLICATIONS

“U.S. Appl. No. 18/203,537, Non Final Office Action mailed Aug. 3, 2023”. cited by applicant

“U.S. Appl. No. 18/203,524, Notice of Allowance mailed Aug. 9, 2023”. cited by applicant

“U.S. Appl. No. 18/203,534, Non Final Office Action mailed Aug. 23, 2023”. cited by applicant

“U.S. Appl. No. 18/203,530, Non Final Office Action mailed Aug. 31, 2023”. cited by applicant

“U.S. Appl. No. 18/203,524, Corrected Notice of Allowability mailed Sep. 7, 2023”, 5 pgs. cited by applicant

“U.S. Appl. No. 18/203,524, Corrected Notice of Allowability mailed Sep. 20, 2023”. cited by applicant

“U.S. Appl. No. 18/203,524, Corrected Notice of Allowability mailed Oct. 5, 2023”. cited by applicant

“U.S. Appl. No. 18/203,537, Response filed Nov. 2, 2023 to Non Final Office Action mailed Aug. 3, 2023”, 17 pgs. cited by applicant

“U.S. Appl. No. 18/203,537, Final Office Action mailed Nov. 20, 2023”, 20 pgs. cited by applicant

“U.S. Appl. No. 18/203,530, Response filed Nov. 21, 2023 to Non Final Office Action mailed Aug. 31, 2023”, 13 pgs. cited by applicant

“U.S. Appl. No. 18/203,534, Response filed Nov. 21, 2023 to Non Final Office Action mailed Aug. 23, 2023”, 13 pgs. cited by applicant

“U.S. Appl. No. 18/203,534, Notice of Allowance mailed Nov. 30, 2023”, 9 pgs. cited by applicant

“U.S. Appl. No. 18/203,537, Examiner Interview Summary mailed Jan. 3, 2024”, 2 pgs. cited by applicant

“U.S. Appl. No. 18/203,530, Notice of Allowance mailed Jan. 12, 2024”, 9 pgs. cited by applicant

“U.S. Appl. No. 18/203,537, Response filed Feb. 20, 2024 to Final Office Action mailed Nov. 20, 2023”, 20 pgs. cited by applicant

“International Application Serial No. PCT US2024 012270, International Search Report mailed Mar. 15, 2024”, 5 pgs. cited by applicant

“International Application Serial No. PCT US2024 012270, Written Opinion mailed Mar. 15, 2024”, 6 pgs. cited by applicant

“International Application Serial No. PCT US2024 012269, International Search Report mailed Mar. 15, 2024”, 4 pgs. cited by applicant

“International Application Serial No. PCT US2024 012269, Written Opinion mailed Mar. 15, 2024”, 7 pgs. cited by applicant

“U.S. Appl. No. 18/203,537, Notice of Allowance mailed Mar. 14, 2024”, 9 pgs. cited by applicant

“U.S. Appl. No. 18/203,537, Corrected Notice of Allowability mailed Jun. 25, 2024”, 3 pgs. cited by applicant

“U.S. Appl. No. 18/783,413, Notice of Allowance mailed Oct. 1, 2024”, 10 pgs. cited by applicant

“U.S. Appl. No. 18/783,416, Non Final Office Action mailed Oct. 1, 2024”, 13 pgs. cited by applicant

“U.S. Appl. No. 18/783,413, Corrected Notice of Allowability mailed Oct. 9, 2024”, 2 pgs. cited by applicant

“U.S. Appl. No. 18/783,419, Non Final Office Action mailed Oct. 18, 2024”, 27 pgs. cited by applicant

“U.S. Appl. No. 18/783,421, Non Final Office Action mailed Nov. 4, 2024”, 20 pgs. cited by applicant

“U.S. Appl. No. 18/783,413, Corrected Notice of Allowability mailed Nov. 6, 2024”, 2 pgs. cited by applicant

Anyword, “Facebook Post Generator”, from Nov. 26, 2022, available at: <https://web.archive.org/web/20221126225805/https://anyword.com/facebook-post-generator> . Accessed Jul. 28, 2023. 6 pages. (Year: 2022), 6 pgs. cited by applicant

Anyword, “Instagram Caption Generator”, archived screenshot from Dec. 4, 2022, available at: <https://web.archive.org/web/20221204212425/https://anyword.com/instagram-caption-generator> . Accessed Jul. 28, 2023., 8 pgs. cited by applicant

Dutta, Debashri, “How to Write a Blog Post Outline in 8 Steps”, <https://web.archive.org/web/20221206180953/https://dmdutta.com/how-to-write-a-blog-post-outline> , posted Oct. 3, 2022, accessed Jul 28, 2023. 13 pages. (Year: 2022), 13 pgs. cited by applicant

Fang, Fei, “A Comprehensive Pipeline for Complex Text-to-Image Synthesis”, *Journal Of Computer Science And Technology*, Springer Singapore, Singapore, vol. 35, No. 3, (May 1, 2020), 522-537. cited by applicant

Gatt, Albert, “Survey of the State of the Art in Natural Language Generation: Core tasks, applications and evaluation”, *Journal of Artificial Intelligence Research*. Jan. 27, 2018;61:65-170. (Year: 2018), 106 pgs. cited by applicant

Kumar, Madhav, “Generative AI and Personalized Video Advertisements”, Available at SSRN 4614118, (2023), 17 pgs. cited by applicant

Lee, Young-Soul, “Automatic Weblog Generation using Mobile Context”, In *Proceedings of the 7th International Conference on Advances in Mobile Computing and Multimedia* Dec. 14, 2009 (pp. 622-626). (Year: 2009), 5 pgs. cited by applicant

Liu, Vivian, “Design guidelines for prompt engineering text-to-image generative models”, In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, (2022), 1-23. cited by applicant

Liu, Vivian, “3DALL-E: Integrating Text-to-Image AI in 3D Design”, *Proceedings of the 2023 ACM Designing Interactive Systems Conference* New York, NY, USA, (Oct. 20, 2022), 1955-1977. cited by applicant

Oppenlaender, Jonas, “Prompt engineering for text-based generative art”, arXiv preprint, (Apr. 20, 2022), 7 pgs. cited by applicant

Oppenlaender, Jonas, “A taxonomy of prompt modifiers for text-to-image generation”, arXiv preprint arXiv:2204.13988v2., (Jul. 31, 2022), 18 pgs. cited by applicant

Smolinski, Pawel, “Towards completely automated advertisement personalization: an integration of generative AI and information systems”, *31st International Conference on Information Systems Development (ISD2023)*, Lisbon, Portugal, (2023), 10 pgs. cited by applicant

Vasquez, Sean, “MelNet: A Generative Model for Audio in the Frequency Domain”, arXiv preprint arXiv: 1906.01083. Jun. 4, 2019. (Year: 2019), 14 pgs. cited by applicant

Vincent, James, “All these images were generated by Google's latest text-to-image AI”, *The Verge*. Vox Media. Retrieved Jul. 28, 2023. 7 pages. (Year: 2022), 8 pgs. cited by applicant

Vivian, Liu, “Opal: Multimodal Image Generation for News Illustration”, *Arxiv.Org*, Cornell University Library, 201 Olin Library Cornell University Ithaca, NY, (Aug. 16, 2022). cited by applicant

Wang, Yuntao, “A Survey on ChatGPT: AI-Generated Contents, Challenges, and Solutions”, *IEEE Open Journal of the Computer Society*, vol. 4 (2023), (2023), pp. 280-302. cited by applicant

Wang et al., “Microblog hashtag generation via encoding conversation contexts.”, arXiv preprint arXiv: 1905.07584. May 18, 2019. (Year: 2019), 10 pgs. cited by applicant

U.S. Appl. No. 18/783,416, Response filed Dec. 23, 2024 to Non Final Office Action mailed Oct. 1, 2024, 29 pgs. cited by applicant
U.S. Appl. No. 18/783,419, Final Office Action mailed Jan. 27, 2025, 33 pgs. cited by applicant
U.S. Appl. No. 18/783,419, Response filed Dec. 23, 2024 to Non Final Office Action mailed Oct. 18, 2024, 18 pgs. cited by applicant
U.S. Appl. No. 18/783,421, Response filed Jan. 23, 2025 to Non Final Office Action mailed Nov. 4, 2024, 17 pgs. cited by applicant

Primary Examiner: Corrielus; Jean M

Attorney, Agent or Firm: Schwegman Lundberg & Woessner, P.A.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation-In-Part of International Application No. PCT/US2024/012269, entitled “Prompting System For Multimodal Personalized Content Generation,” filed on Jan. 19, 2024, which claims the benefit of priority to U.S. patent application Ser. No. 18/203,524, filed May 30, 2023, now U.S. Pat. No. 11,809,688, issued on Nov. 7, 2023, and to U.S. patent application Ser. No. 18/203,530, filed May 30, 2023, now U.S. Pat. No. 11,928,319, issued on Mar. 12, 2024, and to U.S. patent application Ser. No. 18/203,534, filed May 30, 2023, now U.S. Pat. No. 11,922,541, issued on Mar. 5, 2024, and to U.S. patent application Ser. No. 18/203,537, filed May 30, 2023, now U.S. Pat. No. 12,045,735, issued on Jul. 23, 2024, and to U.S. Provisional Patent Application No. 63/444,162, filed Feb. 8, 2023, each of which is herein incorporated by reference in its entirety. This application claims the benefit of U.S. Provisional Patent No. 63/637,254, filed Apr. 22, 2024, and entitled “Multi-Layer Pre-Generated Content;” U.S. Provisional Patent No. 63/611,006, filed Dec. 15, 2023, and entitled “Proactively-Generated Personalized Content Creation;” U.S. Provisional Patent No. 63/637,258, filed Apr. 22, 2024, and entitled “Proactively-Generated Personalized Content Creation;” U.S. Provisional Patent No. 63/637,266, filed Apr. 22, 2024, and entitled “Proactively-Generated Content Creation Based on Tracked Performance;” U.S. Provisional Patent No. 63/644,385, filed May 8, 2024, and entitled “Proactively-Generated Personalized Content Creation;” U.S. Provisional Patent No. 63/637,275, filed Apr. 22, 2024, and entitled “Website Personalization and Interactive Assistant;” and U.S. Provisional Patent No. 63/637,277, filed Apr. 22, 2024, and entitled “Personalized Content Generation.” These provisional applications are herein incorporated by reference in their entirety.

TECHNICAL FIELD

(1) The subject matter disclosed herein generally relates to methods, systems, and machine-readable storage media for generating personalized content for communication messages.

BACKGROUND

(2) Marketers strive to reach the right person at the right time, via the right channel, with the right message to increase product sales. There are automation tools for marketing campaigns that enable marketers to segment audiences for marketing communications. However, the process of creating content for marketing campaigns is slow and expensive. As a result, marketers cannot effectively personalize the messages for all of their audience members at scale.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Various appended drawings illustrate examples of the present disclosure and cannot be considered limiting its scope.
- (2) FIG. 1 is a user interface (UI) for a canvas tool to generate multimodal content, according to some examples.
- (3) FIG. 2 is a screenshot of multimodal content generated by the content-generation tool, according to some examples.
- (4) FIG. 3 is a UI for the management of projects, according to some examples.
- (5) FIG. 4 is an example of a UI to create a new feed of content for a customer journey.
- (6) FIG. 5 is an example of a UI for configuring the goal for the campaign.
- (7) FIG. 6 is an arc UI, according to some examples.
- (8) FIG. 7 illustrates an example of a new, proactively generated blog post.
- (9) FIG. 8 is a UI for presenting the score of an item in the journey, according to some examples.
- (10) FIG. 9 is a UI showing the feed of content, ranked by scores, according to some examples.
- (11) FIG. 10A is an example of a UI to curate content for a customer journey.
- (12) FIG. 10B is a UI to edit multimodal content according to some examples.
- (13) FIG. 11 is an example of a UI to present an audience-centric view of the content for a date-based campaign.
- (14) FIG. 12 is an example of a UI to present an audience-centric view of the content for an always-on customer journey.
- (15) FIG. 13 is an example of a UI to add content to view all the content generated for a in a feed.
- (16) FIG. 14A is an example of a project with multiple feeds.
- (17) FIG. 14B shows an example of a UI with a list of Arcs for the creation and publication of content, according to some examples.
- (18) FIG. 14C illustrates the creation of a new content frame in the arc using the copilot, according to some examples.
- (19) FIG. 15 is an example of two feeds for one or more audiences across one or more channels.
- (20) FIG. 16A illustrates the beginning of the process to create a feed, according to some examples.
- (21) FIG. 16B shows a UI with options to review and modify selections for the feed.
- (22) FIG. 17A illustrates further operations in the process of creating a feed, according to some examples.
- (23) FIGS. 17B-17D illustrate the approval process according to some examples.
- (24) FIG. 18A is an example of a UI that provides performance information for a particular piece of content.
- (25) FIG. 18B is a UI showing arc insights according to some examples.
- (26) FIG. 19 is a diagram of a high-level architecture for implementing embodiments, according to some examples.
- (27) FIG. 20 illustrates details of generation flows, according to some examples.
- (28) FIG. 21 illustrates the process for generating items for user feeds, according to some examples.
- (29) FIG. 22 is a swim chart showing the process for generating content, according to some examples.
- (30) FIG. 23 illustrates the use of a Generative Artificial Intelligence (GAI) model to generate new content, according to some examples.
- (31) FIG. 24 is a flowchart of a method for generating content according to some examples.
- (32) FIG. 25 is a flowchart of a method for generating personalized content according to some examples.
- (33) FIG. 26 is a flowchart of a method for details on generating personalized content according to some examples.

(34) FIG. 27 illustrates the training and use of a machine-learning model, according to some example examples.

(35) FIG. 28 is a flowchart of a method for proactively generating personalized content according to some examples.

(36) FIG. 29 is a flowchart of a method for proactively generating content based on the tracked performance of previous content, according to some examples.

(37) FIG. 30 is a flowchart of a method for generating personalized content, according to some examples.

(38) FIG. 31 is a block diagram illustrating an example of a machine upon or by which one or more example process embodiments described herein may be implemented or controlled.

DETAILED DESCRIPTION

(39) Example methods, systems, and computer programs are directed to the proactive generation of personalized content for use in communication messages. Examples merely typify possible variations. Unless explicitly stated otherwise, components and functions are optional and may be combined or subdivided, and operations may vary in sequence or be combined or subdivided. In the following description, numerous specific details are set forth to provide a thorough understanding of examples. However, it will be evident to one skilled in the art that the present subject matter may be practiced without these specific details.

(40) The present invention relates to a system and method for creating and optimizing highly personalized cross-channel content for marketing campaigns. The system, referred to as Feed Generator (FG) or Content Alchemist, utilizes Generative Artificial Intelligence (GAI) to proactively generate multi-modal content (text copy, images, social media posts, blog posts, emails, audio, videos, etc.) that can be used in marketing campaigns. Additionally, powerful tracking tools are provided to monitor content performance in real-time and proactively suggest optimizations to enhance revenue generation and key performance indicators such as engagement, clicks, and conversions.

(41) The Feed Generator tool enables marketers to create highly personalized content across multiple channels efficiently. The system continuously monitors the performance of the content and provides real-time insights to marketers. These insights include proactive suggestions for optimizations that can drive revenue and improve key performance indicators. Additionally, the Feed Generator integrates with existing Customer data platforms (CDPs) (e.g., Salesforce, Microsoft Dynamics, and ActionIQ), which are software tools that help businesses collect, unify, and activate customer data from a variety of sources, including online and offline channels.

Glossary of Terms

(42) Some of the concepts used for the description of the solution are presented below.

(43) An asset is a product or service offered for sale. Each asset is associated with a group of asset files and additional metadata that provide information about the asset. The files can be of different types, such as text, image, video, etc. Examples of assets include a pair of shoes, a can of soda, a computer, coffee offered by a coffee shop, and income-tax form preparation services. Assets can be associated with a particular project and with a particular catalog.

(44) A catalog is a collection of related assets. Each catalog includes one or more assets.

(45) A prompt is text input used to generate new content. The text input includes instructions in a human language that describe the desired output, which may be multi-modal, such as a piece of text, an image, a video, an audio, a piece of programming code, etc., or a combination thereof.

Types of prompts include a user prompt and a GAI prompt.

(46) A user prompt is a description entered by a user in plain English (or some other language) of the desired content to be generated. The description may be provided as text or in some other form, such as audio, which is then converted to text. In some examples, the content is generated using Generative Artificial Intelligence (GAI) by inputting the user prompt, or an enhanced prompt based on the user prompt, to a GAI tool, also referred to as the GAI model. An example user prompt is

“create an image of product X with a beach background in vibrant colors, where product X takes 30% of the image.”

(47) A GAI prompt is a description in plain English (or some other language) of the desired content to be generated by a GAI tool. The GAI prompt is used as input for the GAI tool, which then generates the desired content. The output of the GAI tool may be multi-modal, such as a piece of text, an image, a video, an audio, a piece of programming code, etc., or a combination thereof.

(48) A canvas is a user interface (UI) that includes one or more fields, including at least a field for entering prompts for the generation of content, a field for showing the generated content, and a field for editing the generated content. The canvas can be associated with a type associated with the desired result, such as an Instagram post, a Facebook post, a Twitter post, a blog post, a Google ad, etc.

(49) A project is an organizational construct to group related work together. It is created by a user with a particular goal (e.g., a marketing goal), such as a Spring Campaign for a new phone release. The project may also be referred to as a campaign. Each project may be associated with one or more canvases and one or more assets.

(50) A publication is a message created for a user, and the message may include one or more blocks of multi-modal content. Examples of publications include a webpage, a text message, an email message, a voice message, an Instagram® message, a blog, an X® message, a Facebook® message, etc.

(51) A canvas template defines the structure of prompts, prompt rules, and other additional context used to generate a canvas.

(52) A publication template defines the structure of prompts, prompt rules, and other additional context used to generate a publication.

(53) A prompt template defines the structure of a GAI prompt used as input to the GAI tool. The prompt template may include fields that can be filled in based on available information to generate the GAI prompt. The prompt template may also include rules for creating the GAI prompt (e.g., include specific text when the recipient is an adult, but do not include the text if the recipient is a minor).

(54) Content refers to elements, also referred to herein as blocks, that can be used in messages targeted to users. The elements are multi-modal because they can be of different types, such as a text block, an image, an audio, a video, an Instagram post, a blog post, an email, a social media post, etc. The content can be generated using Generative Artificial Intelligence (GAI) and edited by users. The generated content is new content generated using the GAI tool. It does not refer to content that is the result of editing or changing existing material (e.g., editing an image to add text to the image is not considered generated new content).

(55) A feed, also referred to herein as a dynamic feed, is a stream of content to help achieve a goal (e.g., driving engagement, increasing revenue, and improving retention across one or more audiences). The content (e.g., text, image, audio, video) of the feed may be multi-modal.

(56) The Feed Generator (FG), also referred to herein as Alchemist, is an application for creating and managing content and feeds.

(57) A journey defines the execution of a plan associated with a project (e.g., send an email to the user about a specific product, wait for three days, and show an advertisement of the product in the user's Instagram feed). The journey includes one or more actions, and each action defines a channel, content to be presented, and one or more rules defining when to present the content. Customer Journeys is one use case for feeds, but the content from a feed can also be used with other communication automation tools.

(58) FIG. 1 is a UI **102** for a canvas tool, also referred to herein simply as canvas, to generate multimodal content, according to some example embodiments. The UI **102** includes a prompt panel **104**, a canvas panel **108**, and a variations panel **106**.

(59) The prompt panel **104** includes a prompt tool **110** for entering a user prompt, which is

descriptive text for the desired results. The prompt tool **110** is more than a simple input field because the prompt tool **110** includes interactive tools for easy entering input.

(60) After input is entered in the prompt tool **110**, the CGS generates results in the variations panel **106** with one or more options that the user can select. The results are referred to herein as variations **112** or outputs. In the illustrated example, the input is for generating an image with specific characteristics, and the variations panel **106** shows several variations **112**. If the user selects one of the variations **112**, the selection is then added to the canvas panel **108**. Additional inputs may be entered in the prompt tool **110**, and variations may be added to the canvas panel **108**.

(61) The Content-Generation System (CGS), also referred to herein as a content-generation tool, is a platform that can generate multiple types of generative content customized for the user and the user's particular environment (e.g., assets, products, services, voice, style, and company of the user). Further, a template-creation tool allows the user to create custom CGS templates to extend and customize the content-generation tool using no-code options that are easy to use. The prompt tool **110** allows the user to express creative ideas naturally and seamlessly integrate with brand assets.

(62) It is noted that examples are presented with reference to marketing-purpose outputs, but the same principles may be used in other environments, such as the generation of documents, contracts, employee newsletters, manuals, instructions, etc. Therefore, the presented examples should not be interpreted as exclusive or limiting but rather illustrative.

(63) GAI can create new content using existing text, audio files, or images. One of the challenges of GAI is that GAI algorithms need a large amount of training data to perform tasks successfully. Another challenge of GAI is unexpected outcomes, so the CGS has to make sure that the generated results are high-quality and safe to present.

(64) FIG. 2 is a screenshot of multimodal content generated by the content-generation tool, according to some example embodiments. In the illustrated example, a created image has been selected and is presented in the canvas panel **108**.

(65) Further, the user has entered the following user prompt in the prompt tool, "Please write a two-page blog post about the benefits of using contract management software. In your post, discuss how it can help create contracts with ease, facilitate collaboration and negotiation, automate contract workflows, manage contracts in one place, and cover opportunities and risk in trends."

(66) The variations panel **106** shows multiple variations **112** for the blog, and the user has selected one of the generated variations to be added to the canvas panel **108**. The different parts of the canvas are editable, including the results and the selected content added to the canvas panel **108**. The process may be repeated, adding new variations (text, image, video) to the canvas. That is, the canvas may be generated through a sequence of content-generation requests until the desired outcome is achieved. This sequence of operations may be saved to create a CGS template, and the user may then use the CGS template in the future to generate similar types of material (e.g., a magazine advertisement, a poster for a conference, multimedia presentation).

(67) The content-generation tool also provides a safety feature to ensure that the content generated is safe, meaning that the user's brand is protected from erroneous content (e.g., incorrect product images) and incorrect grammar or plagiarism. The content-generation tool provides a grammar checker and a plagiarism checker to ensure that the generated content is high-quality and safe. Further, the user can specify what type of content is acceptable and what type is not acceptable.

(68) Furthermore, the CGS includes an authenticity checker for the generated image to ensure the asset is always presented correctly. The CGS provides complete brand control to the user and guarantees that the brand is protected.

(69) FIG. 3 is a UI **302** for project management, according to some example embodiments. The UI **302** lists the projects created by the user. As the example shows, projects may include one or more prompts, which can be of different types (text, image, video, etc.).

(70) In general, people do not want to work with one tool at a time but instead have a tool that

represents the flow of their work. For example, a user may want to create an Instagram ad and would like to create the ad with just one tool instead of having to go to separate tools to create the text and the image. Further, having the text and image correlated provides for a better Instagram ad (e.g., by creating the image based on the selected text). Thus, the projects of the content-generation tool represent the tasks of interest to the user.

(71) Further, the content-generation tool provides a project history (not shown) for each project where the different steps taken to generate items for the project are presented. The user may refer to some of these steps and generate new content, such as by regenerating an image based on a different asset.

(72) FIG. 4 is an example of a UI 402 for creating a customer journey. The UI 402 provides two options: the first option is to create a new journey, and the second option is to connect the FG to an existing journey created with a third-party tool.

(73) By utilizing the Feed Generator, marketers can overcome the limitations of traditional content creation methods and effectively personalize messages for their audience at scale. The system empowers marketers to deliver targeted and engaging content, resulting in improved customer engagement, increased conversions, and, ultimately, higher revenue. Additionally, the FG monitors content performance in real-time and proactively suggests optimizations to help marketers meet their objectives, such as driving more revenue and increasing user engagement, clicks, and conversions.

(74) Up until recently, companies usually had multiple data platforms scattered across the organization. Every department had its own databases for storing customer and transactional records. As a result, it was challenging to gather insights from data in a consolidated manner. However, with the advent of new technologies, these data sources are being integrated and available to be queried using a unified interface. This way, the data is readily available, and users can ask questions and often get a prompt response.

(75) Multiple users may work on communication content for different purposes. The FG operates as the hub for generating content for multiple purposes, unifying content creation while making the content easily accessible.

(76) Modern marketers tend first to identify the channels they want to publish on, then develop a geographical strategy and create audience segments. Further, marketers can leverage databases to generate data-driven content for their brands. Marketers can gather information about their audience and create a specific group of people. Still, then they are limited to using the same small set of content because they are not able to create new content easily. However, with the FG and its unified content hub, FG takes input from the user, e.g., the channel they want to share on, and uses this input to customize the communication. The user can also choose their audience segments, where audiences can be of any size, even having the size of one person. This also allows geography-based customization, as the way to speak to a 25 to 34-year-old person in one country may be different from how someone would speak to that same age group in another country.

(77) Within a single company, there can be multiple brand identities that differ in terms of their tone and voice. For instance, Coke might have a different way of communicating than Sprite. At a broad level, FG provides a layer that enables the user to input data, such as audience or geographic location, or even previous content, which the system can learn from and use to personalize future content. With this layer, the FG can now generate any piece of content quickly and easily, tailored to the user's specific needs. After becoming unconstrained by content, all existing workflows can be reimaged, as this is where the journey begins.

(78) If the user selects the option to create a new plan, another UI, such as the one in FIG. 5, is presented to enable the user to enter parameters for the plan. Suppose the user selects the option to connect the FG to an existing plan. In that case, another UI is presented (not shown) that allows the user to select the existing plan created with other journey orchestration tools like Salesforce®, Journey Builder®, Microsoft Dynamics®, etc.

(79) The use of the FG to create and implement plans has several benefits for the user: 1. Creation of one or more feeds of generated content: Instead of generating content one prompt at a time, users simply curate proactively generated content and personalized feeds available for each audience. This helps marketers quickly pick the “best message” they want to share with their audiences. 2. Omnichannel and multi-modal content: instead of different teams individually creating content for each channel, marketers can visualize and curate the experience for an audience to be consistent across all channels, like emails, SMS, Push Notifications, blogs, and social media. 3. On-brand content creation: instead of customers having to choose between the “brand loop” (a feedback loop that controls the brand image and perception of a company or product) and the “velocity loop” (a feedback loop that controls the speed of content delivery), FG helps users maintain a consistent brand identity, even when users are rapidly creating content across channels. 4. Content Analytics & Insights: FG has a semantic understanding of the content, e.g., the layers of an image, the people, places, and objects in the image, etc. Another example is to identify if there's a tree, a beach, a product, a dog, etc., in the image. FG uses this understanding to get deep insights into why a particular content variant may be performing better. FG then recommends insights—at a journey level and across multiple journeys—and proactive recommendations for generating new content. Some example actions include setting up new journeys, reviewing new suggested content, and replacing existing content.

(80) FIG. 5 is an example of a UI **502** for configuring the campaign goal. The UI **502** includes a field **504** for entering the campaign goal, e.g., selling new Cecilie apple juice.

(81) Once the user enters the goal, the FG presents document **506**, which may be used for the campaign. The documents **506** may be existing documents or documents generated by the FG for the plan. The user may also add, in other following screens (not shown), other details about the campaign, such as the audience, channels selected, performance indicators (KPIs) for the plan, etc.

(82) The FG includes a scaffolding module that facilitates the creation of customer journeys or feeds. In the context of a marketing campaign, scaffolding refers to the initial framework or structure that is set up to support the campaign's goals and objectives. This includes elements such as defining the target audience, identifying the key messages, determining the appropriate channels for communication, and creating a timeline for execution. The scaffolding is the foundation upon which the campaign is built, and it serves as a guide for all the activities that follow. It helps to ensure that the campaign stays on track and that all the different elements work together cohesively to achieve the desired results.

(83) Scaffolding is creating a document or structure that lists all the aspects of the plan. The scaffolding document can be embedded in other journey orchestration tools such as Salesforce Journey Builder, and Microsoft Dynamics. The scaffolding document includes all the necessary aspects of the campaign or journey, provides a comprehensive plan for the marketing activities, and includes the goals for the plan.

(84) The user can edit and refine the plan, and the FG dynamically adjusts the scaffolding for the plan. The FG tool provides cross-section channel content with audience-specific details to enable users to see the messages that each type of audience will receive.

(85) Further, the feed includes all the content created in response to a prompt, a batch request, or some other type of workflow. The FG tool proactively creates, curates, and makes suggestions based on the monitoring of the plan's performance.

(86) The FG tool enables the user to proactively generate content, such as blog posts, emails, social media posts, etc., and the FG tool provides the reason why the content was generated (e.g., based on insight into the performance of the excess income).

(87) Additionally, the FG tool provides insights into the different pieces of content driving use, weeks, conversions, sales, etc., across the different channels. The FG tool uses the insights to proactively generate new content without having to wait for a user request.

(88) One benefit of the FG tool is that it delivers a consistent experience across all channels. In

previous solutions, marketers working through agencies and content for different channels like email, SMS, and social networks would be working in different teams, and it was difficult for all the marketers working together to visualize the end experience for any audience and ensure the user experience was consistent.

(89) Another benefit of the FG tools is that they can generate content quickly because of the understanding of the channels and audiences. Further, the FG tool enables personalization at scale because the FG tool lets marketers create content rapidly, allowing them to create a lot more variance.

(90) FIG. 6 is an arc UI **602**, according to some examples. In literature, a story arc or narrative arc is the chronological structure of a plot in a story. A character arc refers to the journey and transformation undergone by a character throughout a story. The character arc is the evolution of a character's beliefs, personality, or behavior as the character faces challenges and experiences within the narrative.

(91) There can be multiple types of arcs created by marketers in the content generation tool to tell their “story”. One of these types of arcs is a customer journey that a marketer creates for their customers to follow. The customer journey arc relates to the creation and dissemination of content. The arc UI **602** shows an example of a customer journey arc.

(92) The arc includes the following items: The context provided by the user when generating the feed, e.g., the goal of their “story”, the user's products, the user's audiences, and the objectives of each frame of the arc; Creating and curating content from the proactively generated feed. The marketers curate down to what they like and associate the content they like with “frames” in the journey; Publication and activation of content. Once marketers have curated their journeys, they submit the content back into the orchestrator for publishing to end customers; and Content optimization. The content-generation tool surfaces new insights and content recommendations to help marketers create new content that keeps getting better.

(93) The arc UI **602** shows a view of the arc story and provides options to the user, such as adding or removing frames, commenting at the arc level, and seeing insights at the arc level. When the user lands on this page, content may already be available for the user to curate. Some of the icons may have indicators that they need attention (e.g., a yellow alert icon).

(94) At the top of the UI **602**, there is a navigation bar **604** that includes a set of action buttons, including an optimize button, a share button, and a publish button. These buttons enable users to optimize the campaign using performance insights, share it with collaborators, or publish the campaign, respectively.

(95) Below the navigation bar, the main content area displays the project title (e.g., “X100 Tent Arc”), followed by a subtitle or description with additional details about the arc, such as the campaign's purpose and goals (e.g., “Launch and drive sales of the x100 All-Weather Tent”).

(96) The UI **602** includes a tabbed navigation section **606** with tabs labeled “Arc,” “Curate,” and “Pulse,” which allow users to switch between different aspects of the campaign management process. The “Arc” tab shows the view of the customer journey story as described above in [0105]. The “Curate” tab shows the feed as described in [1302]. The “Pulse” tab displays performance metrics.

(97) Below the tabbed navigation section **606**, there is a campaign flow area **608**, which visually represents the sequence of marketing communications planned for the campaign. Each communication is represented by a card **610** that includes a preview of the content, such as “Email 1: Welcome,” “SMS 1: Reminder,” “Email 2: Consideration,” “SMS 2: Reminder2,” and “Email 3: Offer.” Each card **610** is associated with a status indicator, which shows the current state of the communication, such as draft, scheduled, or sent.

(98) To the left of each card **610**, there are action icons that allow users to perform tasks such as editing, duplicating, or deleting the communication. Additionally, a counter may be displayed on the cards, indicating the presence of additional variations of that frame in the arc that are not

currently visible in the interface.

(99) At the bottom of the UI **602**, there is an interactive prompt **612** with a text field that invites users to input additional actions they wish to perform, with the placeholder text “What else do you want to do?”

(100) The UI **602** is designed to be user-friendly and efficient in managing and executing marketing campaigns. It provides a clear visual representation of the campaign flow and easy access to relevant actions and settings.

(101) The project's images include those created based on a prompt entered by the user and those proactively generated by the FG tool. The campaign plan may include information about the plan objective, project KPIs, configured audiences, and communication channels.

(102) Further, the recommendation for new documents includes documents generated by the FG tool to make them readily available for the project. These documents are part of the feed for available communications. The sample recommendations are described below in more detail with reference to FIG. 7. The FG tool generates recommendations for new documents based on user preferences and campaign context. These recommendations may include blog posts or other relevant content. The FG tool proactively generates these recommendations without the need for user prompts. The generated recommendations aim to assist users in their content-generation efforts and enhance the overall performance of their campaigns.

(103) One or more insights provide information about the performance of the project to date. The project provides a view of all cross-channel content so marketers can see what content each of their different audiences will get.

(104) FIG. 7 illustrates an example of a new, proactively generated blog post **702**. The example blog post **702** was proactively generated by the FG without receiving a user prompt; that is, the FG generated the blog post **702** based on the project parameters, including channel and audience parameters. The FG may generate different content for different types of audiences and channels.

(105) The blog post **702** includes an image and text accompanying the image to promote a product. The image and text were generated using the respective GAI tools.

(106) Typically, more than one recommendation is generated by the FG, and the user can click on the scroll arrows on the top right to see additional recommendations. Other options allow the user to publish the blog post **702**, which will send the blog post **702** to the appropriate audience according to the scaffolding schedule, and an option to edit the text or the image (e.g., requesting that a new image be generated).

(107) FIG. 8 is a UI **802** for presenting the score of an item in the journey, according to some examples. The content of the feed is ranked by confidence score, where the higher the confidence score, the greater the probability that the item will be useful and relevant for the campaign.

(108) The UI **802** shows an item for the feed and its confidence score. In this example, the confidence score is **73** for an item that is part of a journey. The UI **802** shows an image associated with an email to be sent as step 2 in the journey and targeted to an audience that includes Electricians in the Bay Area.

(109) The confidence score allows the user to select those items that FG predicts to have a bigger impact on customers, where the impact may be measured, for example, as the number of users that will respond to the email by selecting a link included in the email.

(110) FIG. 9 is a UI **902** showing the feed of content, ranked by scores, according to some examples. The UI **902** shows the content of the feed of available items for communications. The selected option is Best Performing **904**, so the feed shows the available items ranked according to their performance with uses. Performance is a measurement of the response of users to communications. It is based on one or more of the number of views, percentage of users that selected a link for additional information (e.g., click-through rates), percentage of users that bought the product, etc.

(111) In the disclosed system, after a regeneration process, the generated content is presented to the

user. The generated content may include images, blog post recommendations, and social media posts. The generation of this content is based on the information already known by the Alchemist system about the user's company, products, and campaign. The system utilizes past performance data to generate content that is tailored to the user's needs.

(112) The system allows the user to take various actions on the generated content, such as publishing, reviewing, editing, or rejecting it. The user may be presented with multiple options for new blogs or other types of social media content, such as Instagram posts or email recommendations.

(113) The feed allows users to review and take proactive actions on the generated content, such as approving, rejecting, adding to ongoing campaigns, creating variations, editing, and incorporating into the system.

(114) After a goal is provided, the FG quickly curates high-quality, on-brand content in the form of a feed that is ranked based on the objective and personalized across channels and geographies for each audience. The underlying implementation is a semantic graph.

Feed Design & Implementation

(115) A feed of multi-modal content includes emails, images, blog posts, social media posts, text blocks, email subject lines, etc.

(116) Filters based on predicted content performance (some form of a confidence score), channels, audiences, and other attributes like colors.

(117) Dynamic Feed—A feed is a “container” that can be a dynamically generated logical grouping of content based on filters applied. One way to think about the grouping construct is a photo app that supports albums. A photo can be in one or more albums. The album can be created by a human or can be auto-created by AI. Similarly, a piece of content can live in one or more feeds. The feed can be created by a user (by applying filters) or dynamically created by AI.

(118) There are several reasons to create a feed. With feeds, a user can deliver the ‘Best Next Message’ to engage audiences. Feeds are personalized and ranked for each audience to help you connect with your customers. Insights from the performance of live content help you optimize and dynamically adjust the content in real time. Also, with feeds, the user may tell unlimited stories to the different audiences because there is always something new for the audiences to see.

(119) Additionally, feeds tell a cohesive story across all channels. Feeds deliver the best next message a member should see, regardless of channel, and help the user observe a complete view of everything that an audience segment may see across the channels. Feeds can also inspire users because feeds offer visual discovery and brand-personalized inspiration, sparking user interest and creativity through a large amount of content. The user may create a feed with a single prompt and brand assets and then get creative with the possibilities offered by the feed.

(120) After the feed is created, the content can be delivered in many ways, such as: Publish to social media: Users can auto-publish approved content directly to social media channels. Users can use CGS Assistants or plugin feeds in existing tools. Connect to email automation systems: The user can connect email automation systems and in-house marketing platforms to fetch content from a feed directly. For example, during render time, email platforms can fetch and embed a piece of content (like a subject line, pre-header, body copy, image, etc.) for each specific audience. The user may also use the CGS email assistant to manually publish an email into a system like Salesforce Content Builder or Microsoft Dynamics 365. Audience-level personalization on the web: the user can dynamically serve highly personalized content to each visitor on a website by plugging in a feed. On page load, the feed is queried to get the most relevant content for a visitor and display it on the web page. Distribute through Customer Journey Orchestrators.

(121) The user and the CGS can create items and curate the feed. Based on a goal the user specifies, a team of specialized AI assistants generates and curates relevant content for each of the audiences. The user can also manually add or modify content in the feed. Once a campaign is live, the AI assistants proactively create, curate, and suggest content items based on the monitoring of

the performance of existing content.

(122) The user controls what appears in the feed for each audience and can further tailor and make the experience more engaging for each user. The user may add content to a feed or request that the FG generate more content to be added to the feed.

(123) Further, when content is added to the feed, that piece of content is ranked along with others in the feed based on predicted performance. Items in the feed are ranked based on the goal of the feed and the likelihood (e.g., probability) of each item achieving the goal. For example, FG looks at the historical performance of content as a signal when generating new content. In addition, each piece of generated or curated content is assigned a confidence score, and the confidence scores are used to rank the items in the feed.

(124) The confidence score is based on the goal specified when creating the project. This is calculated by evaluating and scoring against a set of criteria (e.g., adherence to brand guidelines, relevancy, match for specified audiences, etc.). However, the user can manually change the ranking of any item to control what audiences will see.

(125) The user can configure feed settings that determine how the feed operates, including generating content, ranking content, etc. For example, the user may configure a default state of new items in the feed (draft or approved); items created as “draft” will require user approval before they are distributed. If this setting is set to “approved,” new items added will immediately be available for consumption.

(126) Another feed parameter is to configure how to handle requests for items that do not exist (default or generate); the user can configure a feed to return a default value or to generate a piece of content on the fly. Another feed parameter is to configure a mode for removing items from the feed (e.g., remove items automatically that are more than a month old, remove items automatically that are not performing well, remove items that have not been used for a predetermined period, etc.).

(127) FIG. 10A is an example of a UI 1002 to curate content for a feed, according to some examples. The UI 1002 shows the content that a specific audience will see. The UI 1002 provides options to curate content for each step in the journey.

(128) In the illustrated example, the UI 1002 presents an email template as part of a customer journey campaign. The UI 1002 includes an email template 1008 titled “Email 1-Variant 1,” which is currently marked with a status indicator “Needs review.” Adjacent to the status indicator is an “Edit” button, allowing the user to make modifications to the email template 1008. Further, the email template 1008 includes a prominent image of an image for the campaign (e.g., a tent in a snowy landscape, with the heading “Introducing Cecilie Juice”) and a subheading, e.g., “The smarter way to a healthy life.” Below the headings, the email template 1008 shows a call-to-action button labeled “Complete your order.”

(129) The body of the email template is presented below in the preview area. In this example, the body of the email contains a personalized greeting text addressing the recipient by name, followed by a marketing message encouraging the recipient to embark on a journey that will transform their coffee experience. The message introduces a new machine to enhance the coffee experience and invites the recipient to include a daily cup of a new beverage in their routine. At the bottom of the email template 1008, there is an interactive query, “What else do you want to do?” with a response field for the recipient to engage with.

(130) To the right of the email template 1008, the UI 1002 includes a secondary display area 1010 that presents alternative generated email options for the email template under the heading “Other layouts.” This secondary display area 1010 features multiple thumbnail previews of different email layouts, each with a unique design and content arrangement. For example, one of the thumbnails is highlighted with a “Selected” label, indicating the current choice of the user. Each thumbnail preview includes a visual representation of the email content, such as images, text, and call-to-action buttons, and is accompanied by interactive elements such as a heart icon for favoriting, a comment icon, and a numerical indicator of the number of comments received.

(131) Clicking on a frame in the secondary display area **1010** would select that specific generation and display it in the preview area on the left. Further, the UI **1002** provides the option for quick preview of other frames, e.g., by clicking on the left and right buttons to cycle through the different frames.

(132) At the top of the user interface, there are additional control options **1012**, including “Optimize,” “Share,” and “Publish” buttons, which provide the user with the ability to optimize the email content, share the template with collaborators, or publish the email as part of the customer journey campaign.

(133) Further, the UI **1002** presents the status of each node and some collaboration features, e.g., who the content is assigned to, who reviewed the content, who approved the content, and who edited the content. The UI **1002** is designed to facilitate the easy selection and editing of email templates within a customer journey, providing a streamlined and intuitive experience for the user to customize and deploy marketing communications.

(134) FIG. **10B** shows a UI **1020** for editing multimodal content according to some examples. The UI **1020** lets marketers use the content-management tool to edit the created content and is designed to facilitate the creation, modification, and publication of multimedia content, such as email marketing materials.

(135) The UI **1020** includes a navigation bar **1022** at the top of the interface, which contains backward **1004** and forward **1006** navigation controls, allowing the user to navigate through different content items. Adjacent to the navigation controls is a content identifier, which provides a title for the current content being edited. The navigation bar **1022** also includes several interactive elements, such as an “Optimize” button, a “Share” button, and a “Publish” button, which allow the user to optimize the content for different platforms, share the content with others for review or collaboration, and publish the content when it is ready, respectively.

(136) The main editing area of the UI **1020** is divided into several sections. The upper section contains an area **1026** with a subject field labeled “Subject” and a preheader field labeled “Preheader,” which allow the user to input and edit the subject line and preheader text of the email content, respectively. In this example, the subject field displays the text “Introducing Cecilie Juice,” and the preheader field contains the text “Check out our new apple juice, the healthiest juice in the market.”

(137) Below the subject and preheader fields is the content editing area **1028**, which displays the multimodal content to be edited. This area includes a visual representation of the email, featuring a prominent image (e.g., a tent in a snowy landscape), followed by a headline (e.g., “Introducing Cecilie Juice,”) a subheadline (e.g., “The smarter way to a healthy life,”) and a call-to-action button labeled “Complete your order.” Additionally, the content editing area **1028** includes a text block with placeholder text for the recipient (e.g., “Dear [Subscriber's Name]”), followed by a message body that can be edited by the user.

(138) The user interface is designed with a clean and intuitive layout, ensuring ease of use for content creators while providing powerful tools for editing and managing multimodal content. The interface elements are arranged to provide a logical workflow from content creation to publication.

(139) FIG. **11** is an example of a UI **1002** to present an audience-centric view of the content for a date-based campaign. The UI **1102** shows the content, organized by audience, that each particular audience will see. The table is organized by calendar date, with each column for a different day, and each row of the table shows the communications planned or delivered for each day for each particular audience.

(140) The system enables users to set up experiments and analyze the results based on different segments. The system provides insights and recommendations for improving audience engagement. Users can compare existing content with new alternatives and make informed decisions based on the recommendations provided. The system also allows users to view recommendations for blog posts, product ads, and other content types. Users can access detailed information about each

recommendation and perform relevant actions.

(141) Further, the FG facilitates an iterative improvement loop where users can configure and select content specifically tailored to different audience segments. By analyzing the performance of different content variations, users can optimize their marketing strategies and enhance audience engagement.

(142) FIG. **12** is an example of a UI **1202** to present an audience-centric view of the content for an always-on customer journey. The information is organized by audience in each row, and the communications that are part of the campaign are shown.

(143) Each communication is represented by a tile **1204**, and selecting a particular tile **1204** (e.g., clicking on the tile **1204** or hovering a mouse over the tile **1204**) will provide additional details about the communication, such as whether the communication is live or being planned, a percentage of users engaging with the communication, the number of messages sent, the dates where the communications are sent, etc.

(144) FIG. **13** is an example of a UI **1302** to add content to view all the content generated for a in a feed. Users can see all content in a feed and review it (for example, to make sure it aligns with their brand guidelines).

(145) At the top of the UI **1302**, there is a navigation bar **1304** that includes a title of the current section (e.g., “x100 Tent Arc”) and a set of action buttons including “Optimize,” “Share,” and “Publish.”

(146) Below the navigation bar **1304**, the main title is prominently displayed (e.g., “x100 Tent Arc”), followed by a subtitle that describes the purpose of the section (e.g., “Launch and drive sales of the x100 All-Weather Tent”) beneath the title. Further, three tab options **1306** “Arc,” “Curate,” and “Pulse” are presented, which allow the user to switch between different functionalities of the UI **1002**.

(147) The central portion of the UI **1002** is for the content management area **1308**, which is divided into a filter bar **1310** and a content display grid **1312**. The filter bar **1310** includes a series of selectable filters such as “All,” “New,” “Draft,” “Approved,” “Favorites,” a filter icon, and dropdown menus for “All channels,” “All steps,” and “All audiences.” These filters enable the user to refine the content displayed in the grid below.

(148) The content display grid **1312** presents a collection of content cards **1314**, each representing a piece of digital content. Each content card **1314** includes a thumbnail image, a content title, and a brief description. Additionally, each content card **1314** has a status indicator (e.g., “New” or “Selected”) and a set of interactive icons for actions like “Add to frame,” “Like,” “Dismiss,” and “More Options.” The content cards **1314** are arranged in a manner that allows the user to browse and select content for further editing or publication easily.

(149) The UI **1302** is designed to provide a user-friendly and efficient environment for content managers to oversee the development and distribution of marketing materials across various stages of a customer journey. The combination of filtering options, content previews, and actionable icons creates a streamlined workflow for users to curate and publish content effectively.

(150) FIG. **14A** is an example of a project with multiple feeds. The UI **1402** is for presenting feeds, and one or more feeds are presented. In the illustrated example, one project includes feed **1404**, titled Winter Campaign, and feed **1406**, titled Black Friday Sale.

(151) Within each feed, details are presented, such as the audience selected (e.g., 18-25-year-old people), channels used (e.g., Instagram, email, SMS), and a list of communications **1408**.

(152) FIG. **14B** shows an example of a UI **1422** with a list of Arcs for the creation and publication of content, according to some examples. The UI **1422** includes a tab named “Arcs,” and the UI **1422** shows the interface when this option is selected. The main area of the UI **1422** is divided into two primary sections: a navigation sidebar **1424** on the left and a content management area **1426** on the right.

(153) The navigation sidebar **1424** includes a search bar for querying content within the system.

Below the search bar, a series of navigation options are provided, including “Home,” “Projects,” “Arcs,” “Templates,” “Brands,” “Audiences,” and “Catalogs.” At the bottom of the navigation sidebar **1424**, additional options for “Settings” and “Notifications” are available. A user profile section displaying the user's name is located at the very bottom of the sidebar.

(154) The content management area **1426** is titled “Arcs” to indicate that this is the selected option in the navigation sidebar **1424** and includes a “Last modified” sorting option and a “Create New Arc” button for initiating the creation of new content arcs.

(155) Below the header, the content management area **1426** displays a series of content blocks **1428**, each representing a different content arc. These content arcs can be of different types including ad campaigns and customer journeys, as indicated by titles such as “Christmas '23—Ad campaign”, “Welcome to Explr Pass—Customer Journey,” and “Winter-themed cart abandonment—Customer Journey.”

(156) Each content block **1428** includes a thumbnail preview of the content, a title, and a brief description. Additionally, some blocks display icons indicating the platforms on which the content will be published, such as “Instagram,” “Email,” or “SMS”. Some content blocks also include a “+2” indicator, suggesting additional content items are associated with that block but not currently visible.

(157) The UI **1422** is designed to provide a user-friendly and efficient workflow for managing various content arcs, from creation to publication, across multiple platforms and campaigns. The arrangement of elements allows for easy navigation, quick access to important functions, and a clear overview of ongoing content strategies.

(158) FIG. **14C** illustrates the creation of a new content frame in the arc using the copilot, according to some examples. The UI **1442** provides options for creating and configuring a content arc.

(159) At the top of the UI **1442**, the title (e.g., “x100 Tent Arc”) is prominently displayed, indicating the current project or campaign focus. Below the title, a descriptive tagline (e.g., “Launch and drive sales of the x100 All-Weather Tent”) is provided to give additional context for the user regarding the purpose of the arc.

(160) The UI **1442** also includes navigation tabs **1444** labeled “Arc,” “Curate,” and “Pulse” situated directly below the title and tagline. These navigation tabs **1444** allow the user to switch between different aspects of the campaign management process.

(161) Below the navigation tabs, a series of content cards **1446** are arranged horizontally, representing different communication moments within the arc. In the illustrated example, the content cards **1446** include several cards associated with the arc, such as “Email 1: Welcome,” “SMS 1: Reminder,” “New Moment,” “Email 2: Consideration,” “SMS 2: Reminder2,” and “Email 3: Offer.” Each content card **1446** may include a visual thumbnail preview of the content and a status indicator, represented by a green checkmark, signifying the completion or active status of the activity in the arc.

(162) A “New Frame” card is selectable to add a new content frame to the arc. If the new-frame option is selected, a prompt box **1448** is presented with the placeholder text “What do you want to create?” to enable the user to input a description or title for the new moment they wish to create.

(163) The prompt box **1448** includes options labeled “Email,” “SMS,” and “Instagram Post” to allow the user to specify the type of content they intend to create for the new moment.

(164) FIG. **15** is an example of two feeds for one or more audiences across one or more channels. The UI **1502** shows feed **1504** and feed **1506**, which have content for one or more audiences across one or more channels.

(165) In the illustrated example, feed **1504** includes two target audiences: California adults 18-25 years old and Washington adults 18-25 years old. Other examples may include more than two audiences configured for the same content. Additionally, feed **1504** also includes communications for different channels, such as Instagram, email, SMS, etc.

(166) A feed can contain content for one or more audiences across one or more channels. For example, one feed could be for email messages and multiple audiences, and another feed can be for web and email communications for multiple audiences.

(167) FIG. **16A** illustrates the beginning of the process to create a feed, according to some examples. In some examples, the content-generation tool identifies suitable options to create the feed and presents them to the user, who may then modify the suggestions.

(168) After the user provides a goal to create a new journey or connects to an existing journey in a third-party system (e.g., Salesforce Journey Builder, Microsoft Dynamics Customer Insights), the content-generation tool sets up the arc using the connected context. The content-generation tool searches for suitable products in connected digital asset management systems identifies different steps in the journey, loads suitable templates, and searches for suitable audiences through connected Customer Data Platforms.

(169) The UI **1602** is designed to guide the user through the initial steps of generating the feed by presenting a series of information fields and options. At the top of the interface, a header section with the title “Generating your feed” is displayed.

(170) The main content area of UI **1602** is organized into a list of sequential tasks, each accompanied by a corresponding icon that suggests the action to be taken or the status of the step. In the illustrated example, the tasks include “Goal,” “Products & Services,” “Templates,” “Audiences,” “Languages,” and “Brand Kit.” Each task includes a circular icon indicating that the content-generation tool is performing the corresponding task.

(171) The “Goal” task includes a subtext, “Understanding your goal,” indicating the purpose or objective of the feed creation process. The “Product & Services” includes a subtext “Finding products to include (searching connected DAMs),” which implies that the content-generation tool is searching for products for the feed.

(172) The “Templates” task includes a subtext, “Picking suitable templates,” indicating that the content-generation tool is looking for a template from various predefined templates for the feed. The “Audiences” task includes a subtext “Identifying audiences (searching connected CDPs),” indicating the search for a target audience.

(173) The “Languages” task includes a subtext, “Configuring languages,” which allows the user to set the languages in which to create content in the feed. The “Brand Kit” task includes a subtext, “Loading your brand kit,” which refers to customizing the feed with brand-specific elements.

(174) At the bottom of the interface, a navigation bar includes two buttons: “Back,” which allows the user to navigate to the previous step in the process, and “Next,” which enables progression to the subsequent step. Additionally, a “Cancel” button is provided, offering the user the option to terminate the feed creation process at any point.

(175) The interface is designed with a clean and straightforward layout, utilizing a color scheme that contrasts the text and icons against the background for clear visibility. The use of icons next to each step provides a visual cue to the user, enhancing the user experience by making the process more intuitive.

(176) FIG. **16B** shows a UI **1620** with options to review and modify feed selections created for a customer journey. The content-generation tool presents the items identified for the feed, and the UI **1620** enables the user to review or modify the suggestions for the marketing campaign feed. The UI **1620**'s title, “Review,” is displayed at the top, indicating that the user is in the review stage of the campaign setup process.

(177) The UI presents the same tasks described above with reference to FIG. **16A** and the values proposed by the content-generation tool. In the illustrated example, the “Goal” task displays the campaign's objective, which in this case is “Launch and drive sales of the X100 All-Weather Tent.”

(178) The “Product & Services” task includes two versions of the product “X100 Tent” and allows the user to review and potentially modify the products or services being promoted in the campaign.

(179) The “Journey Activities” task includes entries such as “Email 1—Welcome email” with

details like the goal “Introduce X100 Tent” and a brief tag description “Say hello to your new Travel companion.” Additionally, a visual representation of the email layout is provided. Another entry, “Push Notification,” is shown with the goal “Introduce X100 Tent” and the hashtag “#EmbraceTheElements.” A pull-down menu labeled “See all” allows the user to expand the list to view additional activities.

(180) The “Audiences” task shows two suggested audience segments: “Millennials” and “Young Adults,” indicating the demographic groups the campaign is intended to reach. The “Languages” suggests “English” and “Italian” as languages for the communications. The “Brand Kit” task suggests a kit labeled “Summer rand Kit,” which refers to a set of branding materials or themes to be used in the campaign.

(181) At the bottom of UI **1620**, navigation and action buttons are provided: a “Back” button, a “Cancel” button, and a “Generate” button, which allows the user to proceed with the campaign generation based on the selections.

(182) FIG. **17A** illustrates further operations in the process of creating a feed, according to some examples. The UI **1702** is used to configure the channels to reach the selected audiences. The channels are configurable by the user. In some examples, the available channels include one or more of email, Instagram® post, Facebook® ad, Google® ad, X® (previously Twitter) post, LinkedIn® post, SMS message, and YouTube® ad. The user may select any channel by selecting the corresponding box in the UI **1702**.

(183) The FG automatically selects products from the user library that match the objective. Additionally, UI **1704** is presented to confirm or change the selected products and brand profile.

(184) FIGS. **17B-17D** illustrate the approval process according to some examples. FIG. **17B** shows part of a UI **1712** for previewing the generated content and provides an option to publish. Once the user is done with the content, the user can select the option “Approve” from a drop-down menu **1714** and change the status of the content.

(185) In the illustrated example, the UI **1712** interface includes a dropdown menu **1716** titled “Email 2—Lorem ipsum dolor sit amet” to select from the different pieces of content available in the feed.

(186) The content area **1718** displays an email composition pane with a subject line that reads “Introducing Cécilé Juice” and a preheader text stating, “Check out our new apple juice, the healthiest juice in the market.” The content is presented below the preheader. In this illustration, only a part of the complete UI is presented, and the user may scroll down the page to see all the content.

(187) At the top right corner of the UI **1712**, there are several action buttons and indicators. The “Optimize” button is an option to improve the email's content or deliverability. Adjacent to this is the “Share” button, which allows the user to share the email draft with others for collaboration or review. The drop-down menu **1714** “Needs review” is available to approve the content, and a “Publish” button is provided for sending the content to the email marketing or customer journey orchestration system.

(188) FIG. **17C** shows the UI **1712**, where the user selects the option “Approved” in the drop-down menu **1714**. The opposite operation is also possible, where the user may change the status from “Approved” to “Needs review.”

(189) FIG. **17D** shows a UI **1712** after the user has approved the content, showing the drop-down menu **1714** with the selected option of “Approved.” Once the content is approved, the campaign will start.

(190) FIG. **18A** is an example of a UI **1802** that provides performance information for a particular piece of content. The CGS identifies key visual elements, such as backgrounds and colors, within the images and utilizes a machine-learning model to compare the performance of these features. Based on the comparison, the CGS then identifies differences between the images, allowing for the generation of performance insights and recommendations for image replacement.

(191) The illustrated example shows a predictive uplift in engagement when using blue backgrounds, which is because images with blue backgrounds get better responses from customers.

(192) The CGS system provides performance insights on existing content. For example, if an email has already been sent out as part of a campaign, the system can analyze the performance of the included images and text. This goes beyond the current practice of A/B testing subject lines, as it provides insights into the specific text or body copy. The system can identify which phrases or images are performing better, such as the inclusion of a person's name in the subject line, leading to more opens, or the use of an image with a blue background in a nature scene, resulting in better performance. These insights offer a deeper understanding of content performance.

(193) In one example, the CGS first identifies the key visual elements present in the images, such as objects, backgrounds, and colors. This identification process involves analyzing the image data and extracting relevant features. The system may utilize image recognition algorithms and deep learning techniques to identify and categorize these visual elements accurately.

(194) Once the key visual elements are identified, the CGS creates a machine-learning model that compares the features of the images. This model is trained using a dataset of images with known performance metrics, allowing it to learn patterns and correlations between the visual elements and content performance. The model can then be used to predict the performance of new images based on their visual features.

(195) The CGS provides a novel approach to analyzing and comparing images for content performance insights. By identifying key visual elements and utilizing a machine learning model, the CGS enables content creators to optimize their content and improve engagement. The CGS's predictive capabilities offer valuable recommendations for image replacement, leading to enhanced content performance.

(196) The illustrated example shows an insight for comparing several images that have been presented to users, where the different images have different color backgrounds. The insight indicates that blue backgrounds are performing better in the period being tested.

(197) The CGS actively monitors the performance of content in relation to the marketer's objectives, campaign journey, and key performance indicators (KPIs) and generates content suggestions proactively, even before a specific prompt is given by the user. The CGS then presents these suggestions to the marketer, who can either approve or reject them or make quick edits before incorporating them into the campaign.

(198) FIG. 18B is a UI 1812 showing arc insights according to some examples. The UI 1812 is designed to provide users with an intuitive and comprehensive overview of various marketing campaign elements and their performance metrics.

(199) The UI 1812 includes a header section labeled "x100 Tent Arc," which indicates the current marketing campaign being viewed. Below the header, there are tabs 1818 labeled "Arc," "Curate," and "Monitor," which allow users to navigate between different aspects of the campaign management system.

(200) The main area of the interface displays a series of cards 1820, each representing a different stage of the marketing campaign, such as "Awareness," "Reminder SMS," "Consideration," and "Offer SMS." Each card includes a visual preview of the campaign content, a status indicator, and additional information or actions that can be taken, such as viewing more details or adding new content.

(201) The UI 1812 includes a navigation bar at the top with an "Optimize" option 1814, a "Share" option, and a "Publish" option. When the user selects the "Optimize" option 1814, the panel 1816 is presented with insights information. The user can select and apply any of the insights.

(202) The panel 1816 includes a numerical score (e.g., 88) indicating the Arc performance of the campaign. Below the performance score, there is an "Insights" section providing actionable suggestions to improve the campaign. Examples of insights include demographic performance notes and content optimization recommendations.

(203) Additionally, the panel **1816** contains a “Brand” section that outlines the values and tones most prominent in the content, with indicators such as “Empowerment” and “Innovation” and corresponding progress bars to visualize the extent to which these values are represented.

(204) At the bottom of the UI **1812**, there is an interactive element **1822** prompting the user with the question “What else do you want to do?” suggesting further actions or navigation paths the user may wish to explore.

(205) FIG. **19** is a diagram of a high-level architecture for implementing embodiments, according to some examples. In some examples, the CGS includes several layers for implementation, including apps **1902**, API **1904**, and workers **1906**. The apps **1902** provide customers with the tools to access the CGS. The core app is the program that creates content.

(206) Different types of applications allow the user to create content specifically customized and optimized for a particular output channel. To achieve this, specialized apps are provided, e.g., the strategy for email campaigns is different from the strategy for ad campaigns.

(207) The alchemist app indicates any app that may use the Feed Generator, **1910**. For example, the Arc app for users creating a customer journey to drive sales of a particular product, such as Cicilie apple juice. The ads app **1,912** indicates a different application of the Feed Generator that specializes in generating advertisements.

(208) The API **1904** refers to the programmatic access to configure the journey generation. The API **1904** includes goal **1914**, feed **1916**, brand kit **1918**, and audience **1920**. The goal **1914** is used to define a specific goal in mind for the feed **1916**, e.g., to understand both the brand and the audience they are trying to reach. In some examples, a dynamic feed definition takes four inputs: goal for the activity, like a campaign that drives the feed content generation; channel information (like email, social media, etc.); audience description for which the content generation is targeted; and information retrieval sources from the grounding of the generation will happen (like product information and brand assets).

(209) The brand kit **1918** refers to the personalization of the content based on the brand. The FG ensures that the content looks and feels as if it is coming from the brand by using the brand kit, which includes all the characteristics that have been defined for the brand, such as image styles, text tones, voicemail, etc. Additionally, the feed **1916** utilizes information from the audience **1920** to determine the target of the content. The feed **1916** is then generated based on these parameters.

(210) AI agents, also referred to as workers **1906**, are used to generate and curate content. In one example, the workers **1906** are built with orchestration on SOTA LLMs, and the image generation models continually use feedback based on content insights to generate new content. The content is built with meta prompts and agentic workflows to drive content for a feed. The meta prompts include plans and guidelines to generate new content, like an email journey for a new customer onboarding.

(211) The feed generator **1922** generates content for the user. The feed generator relies on the planner **1924** to get the information for generating the content. The planner **1924** acts as an assistant that ensures all the necessary information to create content is available.

(212) The planner **1924** has a deep understanding of various channels, such as email and ads, and their respective requirements. Additionally, the planner **1924** takes into consideration the audience and the brand attributes. The planner **1924** is able to access databases for product details and pricing information to include in communications.

(213) The planner **1924** acts as an assistant to help plan marketing campaigns. The user specifies which audiences to target and what product to promote, and the planner **1924** creates a plan for the user. To do this, the planner **1924** gathers insights and information, e.g., what works best for young adults on Instagram? and then may gather data and images to help optimize the campaign. The planner **1924** acts as an orchestrator, coordinating all the different elements of the campaign to ensure maximum success.

(214) It is possible to execute a multiple-channel campaign. For instance, to use both social media

and email marketing. In this case, the planner **1924** would create a feed **1926** for email and a feed **1930** for social media.

(215) FIG. **20** illustrates details of generation flows, according to some examples. To generate personalized content, The CGS incorporates a robust semantic content graph, agent operators, and channel-based feeds in a Generative AI stack.

(216) The CGS has three components for generative workflows: 1. A semantic content graph that stores already generated content or helps with the generation of new content; 2. A dynamic feed definition that takes a goal, audience, brand information, insights, and channel as inputs and works on top of the semantic graph to generate new content for the channel based on the customer's needs; and 3. AI Agents (workers) that generate and curate content for each of the feeds based on feed definition and content insights.

(217) The planner **1924** pulls from existing data sources and insights **2002** to create content for the campaign, e.g., analyzing the performance of previous campaigns to create and select new content. For example, if the user has a collection of food images and user searches for “pasta,” the semantic memory **2006** recognizes that pasta is a type of food and extracts food images from the existing sources and insights **2002**. The semantic memory **2006** semantically understands the relationship between pasta and food and makes the necessary connections.

(218) The planner **1924** gathers brand information from both procedural memory **2004** and semantic memory **2006**. The procedural memory **2004** is responsible for creating the plan for content creation, and the semantic memory **2006** determines the meaning of different content. The procedural memory **2004** provides the generated content based on signal. Further, the planner **1924** stores emerging learning data in semantic memory, such as new insights on the dynamic audience and a new journey plan based on insights.

(219) When a query **2010** for the dynamic feed is received, the semantic query with extracted feature labels and filters is sent to the planner **1924**. The query **2010** includes user context. For example, the user may request to get all the content for an email or other content that showcases a certain product. The planner **1924** then creates the feed.

(220) The steps of a plan execution **2012** include the steps for executing a customer journey, which may include multiple steps. For instance, in an abandoned cart scenario, the first step would be to remind the person that they forgot to check out, the second step would be to send a notification after three days offering a ten percent discount, and the third step would be to send a note after another three days informing the customer that supplies are running low and that if they want to purchase the item, they should check out now. This is the customer journey that makes use of the dynamic feed.

(221) The following points provide a high-level overview of the operational process of Retrieval Augmented Planning and Generation, utilizing an agent framework: 1. Receive a feed creation request articulated in natural language. 2. The CGS retrieves the necessary tools and metadata information from the semantic store, which are required for the subsequent plan formulation. 3. The CGS creates a plan of action to accomplish the task at hand. This is done by employing a generative AI model, into which the pertinent tool and metadata information are input. 4. The plan is executed by activating the necessary tools, carrying out the specified intermediate steps, and utilizing the context information retrieved from both the application and the semantic store. 5. As the process proceeds, intermediate steps and results are recorded and stored back into the semantic store. 6. If the CGS requires further clarification to execute a query or to make progress, it seeks this from the user. This newly obtained information is then stored back into the semantic storage system. 7. Finally, once the result is generated, the CGS stores the result in the semantic storage and simultaneously returns the result to the application as part of the ongoing session. The output can take various forms, including textual content or multi-modal content such as graphs, plots, or tables.

(222) The following are some design considerations for storage and retrieval: Relevance Lookup:

The semantic storage system must be designed to allow the retrieval of contextual information, which will be input into the context window of a Language Learning Model (LLM). This feature yields near-text results, as the quality of the output is significantly influenced by this factor. Coherence: a method is established to synchronize this store with a reliable source, such as application storage, ensuring that the data remains consistent across both. In one example, a look-aside pattern is used, but other methods may be used to continually monitor and maintain the synchronization between the semantic store and the original storage. Compliance & Security: When dealing with customer data that will be indexed in a CGS abstracted semantic storage system, it is important to adhere to all relevant compliance and security requirements. Considerations include employing customer-managed keys (CMK) and end-to-end (E2E) encryption. Efficiency: To keep the cost of embedding re-computation low for lookups, the storage and retrieval system must be designed with efficiency in mind.

(223) Regarding security, since the content might contain Personally Identifiable Information (PII) and sensitive information, the proper compliance requirements are implemented to protect customer privacy. One of the considerations is to look at where the store is located and how retrieval happens from the store.

(224) Right prompt templates with the ground truth are implemented because it is important to make sure the model does not hallucinate, or if the model hallucinates, to detect and possibly correct hallucinations.

(225) The system performs periodic performance evaluation and fact-checking. Agents can get into an error loop or be faulty with incorrect information. This might require evaluating and getting the system out of fault by reconstructing the prompt automatically or by requesting user intervention.

(226) FIG. 21 illustrates the process for generating items for user feeds, according to some examples. While the various operations in this flowchart are presented and described sequentially, one of ordinary skill will appreciate that some or all of the operations may be executed in a different order, combined, omitted, or executed in parallel.

(227) The apps **1902** interface with an API head **2106** (e.g., API interface) to submit requests. The apps **1902** include the core app **1,908**, the alchemist app **1,910**, the app **1,912**, a Microsoft Dynamics app **2,102**, and a Salesforce app **2,104** that interfaces with a Salesforce system to gather information about the user's campaigns.

(228) The API head **2106** sends requests to the planner **1924** to plan the content to be created. The planner **1924** sends requests to generate content to the feed generator **1922**. Further, the API head **2106** may also send feed-create requests **2108** to the feed generator **1922**.

(229) The feed generator **1922** sends feed-create requests **2114** to a feed scheduler **2118** that manages the scheduling of request processing in the CGS. Further, a feed analytics processor **2116** monitors the performance of the generated content.

(230) FIG. 22 is a swim chart showing the process for generating content, according to some examples. The swim chart describes the interactions among the app **1,902**, the API head **2206**, the planner **1924**, the feed executor **2202**, a queue **2204** (holds creation requests), and the scheduler **2118**.

(231) The app **1,902** sends a call to the API head **2206** to generate ideas with a goal specified in natural language and optional information about assets and channels.

(232) The API head **2206** then sends a request to the planner **1924** to, given a goal and profile, generate content definitions, one per channel, with audience and product information. The feed supports multiple channels, audiences, and products.

(233) The planner **1924** uses LLMs and agents to develop a list of content for the feed. The planner **1924** then returns feed definitions to the API head **2206**, and the API head **2206** persists (e.g., stores) the feed definitions in a database.

(234) The API head **2206** sends a message to be placed in the queue **2204** for processing, the message including a request to generate new feed items for the feed. Further, the API head **2206**

sends a scheduling request to the feed scheduler **2118** for each definition to refresh the feed content periodically (e.g., every 24 hours). After the content is generated, the API head **2206** returns a list of feeds generated to the app **1902**.

(235) The feed executor **2202** reads messages from the queue **2204** and, based on the feed definition, generates feed items for each message in the queue **2204**. The feed executor **2202** then sends the feed definitions to the API head **2206**.

(236) The app **1,902** consumes feed items by sending requests to the API head **2206** for feed items for a given feed, and the API head **2206** returns the feed items already generated for the specified feed.

(237) FIG. **23** illustrates the use of a Generative Artificial Intelligence (GAI) model **2312** to generate new content, according to some examples. GAI is a type of AI that can generate new content, such as images, text, video, or audio. The GAI model **2312** is trained on large datasets of data and uses this data to learn the patterns and relationships between different elements of the data. There are several types of GAI models, such as Generative Adversarial networks (GANs), Variational Autoencoders (VAEs), and Autoregressive models.

(238) The GAI models generate items of different types, such as GAI models for creating text (e.g., GPT-4, Pathways Language Model 2 (PaLM 2), LaMDA), images (e.g., DALL-E 2, Stable Diffusion), videos (Runway Gen-2, Stable Diffusion Video), audio (e.g., Google MusicLM, Stable Audio), etc.

(239) Often, the companies that create the GAI models make the GAI models available to users who can apply them to generate the desired content based on a GAI prompt **2310** provided to the GAI model **2312**. Users can utilize the GAI model **2312** as provided by the vendor or can optionally fine-tune **2314** the GAI model **2312** with their user data to adjust the parameters of the GAI model **2312** in order to improve performance on a specific task or domain.

(240) In some examples, fine-tuning the GAI model **2312** includes the following operations: 1. Collect user data: Gather a collection of user data that is relevant to the target task or domain. This data could include text, images, audio, or other types of data; 2. Label the data: if the task requires supervised learning, the user data is labeled with the correct outputs; 3. Select a fine-tuning method. Some of the methods for fine-tuning GAI models include Full fine-tuning, Few-shot fine-tuning, and Prompt-based fine-tuning; 4. Train the GAI model **2312**: Perform incremental training of the tune **2314** using the selected fine-tuning method and 5. Optionally, evaluate the performance of the fine-tuned model on a held-out dataset.

(241) The GAI model **2312** can be used to generate new content based on the GAI prompt **2310** used as input, and the GAI model **2312** creates a newly generated item **2316** as output.

(242) The GAI prompt **2310** is a piece of text or code that is used to instruct the GAI model **2312** towards generating a desired output (e.g., generated item **2316**). The GAI prompt **2310** provides context, instructions, and expectations for the output. The newly generated item **2316** may be multi-modal, such as a piece of text, an image, a video, an audio, a piece of programming code, etc., or a combination thereof.

(243) Prompt engineering is the process of designing and crafting prompts to effectively instruct and guide a GAI model toward generating desired outputs. It involves selecting and structuring the text that forms the GAI prompt **2310** input to the GAI model **2312**, ensuring that the GAI prompt **2310** accurately conveys the task, context, and desired style of the output.

(244) A prompt generator **2308** is a computer program that generates the GAI prompt **2310**. There are several ways to generate the GAI prompt **2310**. In one example, the prompt generator **2308** may use a user prompt **2306** entered by the user in plain language as the GAI prompt **2310**. In other examples, the prompt generator **2308** creates the GAI prompt **2310** without having a user prompt **2306**, such as by using a static pre-generated prompt based on the desired output.

(245) In other examples, the prompt generator **2308** uses a prompt template **2302** to generate the GAI prompt **2310**. The prompt template **2302** defines the structure of the GAI prompt **2310** and

may include fields that may be filled in based on available information to generate the GAI prompt, such as user data **2304** or user prompt **2306**. The prompt template may also include rules for creating the GAI prompt (e.g., include specific text when the recipient resides in California, but do not include the text if the recipient does not reside in California). In other examples, the prompt generator **2308** uses heuristics codified into a computer program to generate the GAI prompt **2310**. (246) An example template for generating a prompt to create an image for a marketing campaign is as follows:

(247) You are an AI Assistant that will help create prompts that can be used to generate images with Dall-e and stable diffusion.

(248) The user can provide information as input for generation: 1. Product or service information (e.g., selling t-shirts) 2. Goal—Example—I want to run a promotion for Halloween 3. Audience Information—Location, Gender, Demographics, Age group 4. Channel—Email, Instagram, TikTok, etc.

(249) Use the above information in the following way to generate the prompt.

(250) Product Information:

(251) Clearly highlight the main product.

(252) Example: “A t-shirt”

(253) Goal:

(254) Incorporate the objective or theme of the promotion.

(255) Example: “with a unique Halloween-themed design”

(256) Audience Information:

(257) Add elements that resonate with the target audience's location, gender, demographics, and age group. Ensure the depiction is inclusive and diverse.

(258) Example (based on hypothetical audience info): “being worn by a young male from Italy in his 20s”

(259) Channel:

(260) Tailor the style and format based on the channel you're targeting. For instance:

(261) Email: A more detailed and descriptive scene as it's viewed on larger screens.

(262) Instagram: Focus on aesthetics and vibrant visuals.

(263) TikTok: Dynamic and engaging visuals, perhaps hinting at movement or a narrative.

(264) Example (for Instagram): “The setting is a moonlit night in an Italian piazza, where the young man joins friends for a Halloween celebration. The atmosphere is festive with jack-o'-lanterns, cobblestone streets, and ancient buildings in the background. The image is vibrant and aesthetically pleasing, perfect for an Instagram post.”

(265) Final Prompt:

(266) “A t-shirt with a unique Halloween-themed design being worn by a young male from Italy in his 20s. The setting is a moonlit night in an Italian piazza, where the young man joins friends for a Halloween celebration. The atmosphere is festive with jack-o'-lanterns, cobblestone streets, and ancient buildings in the background. The image is vibrant and aesthetically pleasing, perfect for an Instagram post.”

(267) Remember, this is just one example. Depending on the specifics of the audience information and the chosen channel, the prompt can be tailored differently. The idea is to incorporate all the provided details into a cohesive and engaging narrative for the image.

(268) Creating a good prompt structure for generating images with models like DALL.Math. E or Stable Diffusion requires a balance of specificity and creativity. Here's a breakdown of the essential elements and how you can structure them: 1. ****Type of Image:**** Start by specifying the format or medium of the image. For example: Photo Oil painting Watercolor painting Illustration Cartoon Drawing Vector Render 2. ****Main Subject:**** Clearly identify the primary focus or subject of the image. Be as specific as you can. Instead of “bird,” say “a crimson-feathered parrot.” 3. ****Action or State:**** Describe what the subject is doing or the state it's in. For instance, “soaring through a

clear blue sky” or “perched on a wooden branch.” 4. ****Setting or Background:**** Provide context by describing the environment or setting. This could be “a bustling city square,” “a serene lakeside,” or “a starry night.” 5. ****Additional Details:**** Enhance the scene with additional elements or props. These can be objects, secondary characters, or even weather conditions. For example, “with children flying kites” or “as rain gently falls.” 6. ****Mood or Atmosphere:**** If you have a particular mood or tone in mind, specify it. Words like “sunset,” “dreamy,” “nostalgic,” or “futuristic” can guide the generation. 7. ****Diversity and Representation:**** Especially for images with people, ensure that you specify diverse gender, racial, and other representations to avoid unintentional biases. 8. ****Perspective and Orientation:**** Mention if you want a specific viewpoint like “bird’s eye view” or “close-up.” Also, if the image should be “wide” (landscape) or “tall” (portrait), include that.

(269) Here’s a structured example: “Photo of a diverse group of young people sitting around a campfire in a dense forest at twilight, laughing and sharing stories, with sparks from the fire rising into the starry night.”

(270) Remember, while it’s essential to be specific to guide the model, leaving a bit of room for creativity can lead to surprising and delightful results. Adjust the level of detail based on the desired outcome.

(271) Targeting imagery based on audience demographics like location, gender, and age can enhance the effectiveness of the visual content. Here’s how you can tailor image prompts to appeal to specific audience attributes:

- **Location:**** ****Cultural References:**** Incorporate elements, symbols, or landmarks that resonate with people from that specific region. ****Landscapes & Settings:**** Use typical settings from the location, such as cityscapes, countryside, beaches, or mountains. ****Color Palette:**** Some cultures have color preferences based on traditions, seasons, or festivals.
- **Gender:**** ****Inclusivity:**** Ensure representation of all genders. Avoid stereotypes; instead, focus on universal themes or elements that resonate with the target gender. ****Themes & Motifs:**** Depending on research and insights, you might incorporate certain themes, patterns, or motifs that appeal to a specific gender.
- **Age:**** ****Children (below 12 years):**** ****Style:**** Cartoons, animations, or illustrations with bright colors. ****Elements:**** Friendly animals, magical creatures, superheroes, and playful settings. ****Teens (13-19 years):**** ****Style:**** Trendy illustrations, digital art, or photographs. ****Elements:**** Pop culture references, technology, hobbies, and youth-centric activities. ****Adults (20-60 years):**** ****Style:**** Realistic photos, classic paintings, or mature illustrations. ****Elements:**** Daily life, work, family, nature, hobbies, and contemporary themes. ****Seniors (60+ years):**** ****Style:**** Vintage or nostalgic imagery, calming watercolors, or serene photos. ****Elements:**** Nature, landscapes, nostalgia, classical art, and leisure activities.
- **Interests & Hobbies:**** If you have information on the target audience’s interests, incorporate elements that resonate with those hobbies or passions.
- **Emotional Appeal:**** Think about the emotion you want to evoke. Children might respond to joy and wonder, teens to aspiration and adventure, adults to connection and nostalgia, and seniors to tranquility and reflection.
- **Narrative & Storytelling:**** Craft a narrative that resonates with the target group. Kids might enjoy a story of adventure, teens a coming-of-age tale, adults a journey of discovery, and seniors a trip down memory lane.

(272) When constructing your prompt, always combine these tailored elements to create a cohesive and appealing image. For instance: “Cartoon illustration of a magical forest with bright-colored trees, where friendly animals host a picnic for children, with a backdrop of a famous landmark from [Location].”

(273) Remember, while these are general guidelines, individual preferences can vary widely. It’s essential to understand your specific audience and adjust the attributes accordingly.

(274) Incorporating channel-specific information into your image prompts is crucial because each platform has its own audience expectations, content style, and purpose. Here’s how you can tailor your prompts based on popular social media channels:

- **Instagram:**** ****Purpose:**** Primarily

visual content; used for personal sharing, lifestyle branding, and visual storytelling. ****Prompt Attributes:**** Focus on aesthetics, vibrant colors, and visually pleasing compositions. ****Example:**** “Photo of a minimalist workspace with pastel-colored stationery, a succulent plant, and a morning coffee cup, suitable for a lifestyle influencer on Instagram.” 2. ****TikTok:**** ****Purpose:**** Short-form video platform known for fun, entertaining, and sometimes educational content. ****Prompt Attributes:**** Emphasize dynamic, youthful, and trendy elements. Consider popular challenges, dances, or meme formats. ****Example:**** “Illustration of a diverse group of young people dancing in trendy outfits to a popular song, with vibrant neon lights in the background, perfect for a TikTok dance challenge.” 3. ****LinkedIn:**** ****Purpose:**** Professional networking and industry news. ****Prompt Attributes:**** Prioritize professional, clean, and formal visuals. Showcase industry-related themes, corporate environments, and professional individuals. ****Example:**** “Photo of a diverse group of professionals in a modern office boardroom, discussing a project with charts and graphs projected on a screen, suitable for a LinkedIn post on teamwork.” 4. ****Twitter:**** ****Purpose:**** News, updates, personal opinions, and quick information sharing. ****Prompt Attributes:**** Go for relevant, timely, and sometimes witty or satirical visuals. Infographics or concise visuals work well. ****Example:**** “Illustration of a bird (representing Twitter) perched on a globe, tweeting out news headlines, ideal for a tweet about current global events.” 5. ****Facebook:**** ****Purpose:**** Personal updates, news sharing, and community interactions. ****Prompt Attributes:**** Focus on relatable, family-friendly, and community-centric visuals. ****Example:**** “Photo of a community event in a local park, with families enjoying a barbecue and kids playing games, fitting for a Facebook community group post.” 6. ****Pinterest:**** ****Purpose:**** Inspiration, DIYs, recipes, and mood boards. ****Prompt Attributes:**** Emphasize creativity, inspiration, and how-to visuals. ****Example:**** “Illustration of a step-by-step DIY craft project, showcasing materials, and the final product, perfect for a Pinterest pin.” 7. ****YouTube:**** ****Purpose:**** Long-form video content ranging from entertainment to tutorials. ****Prompt Attributes:**** Create thumbnails or visuals that encapsulate the video's theme and are attention-grabbing. ****Example:**** “Illustration of a person in front of a computer, with graphics of video editing tools around them, suitable for a YouTube tutorial on video editing.”

(275) When crafting prompts for each channel, always consider the platform's primary audience, content format, and purpose. Tailor your visuals to resonate with the users of that specific platform and align with the content type they expect to see.

(276) Using the provided audience segmentation types, you can craft a detailed and targeted image prompt. Let's break down how to incorporate each segmentation type into the prompt: 1.

****Demographic Segmentation:**** Utilize outward-facing attributes to describe individuals or groups in the image. ****Example (based on hypothetical data):**** “A middle-aged married woman”

2. ****Behavioral Segmentation or Persona:**** Integrate actions or habits that resonate with the audience's behavior. ****Example (based on hypothetical data):**** “who frequently shops online, actively engages on social media, and enjoys mobile gaming on her tablet” 3. ****Psychographic or Attitudinal Segmentation:**** Infuse elements that align with the audience's interests, preferences, and values. ****Example (based on hypothetical data):**** “She has a preference for eco-friendly brands, enjoys mystery movies, and values sustainability and wellness.” 4. ****Geographic Segmentation:**** Set the scene in a location that matches the audience's geographic region.

****Example (based on hypothetical data):**** “She is situated in a cozy apartment in downtown Toronto, Canada.” ****Final Prompt:**** “A middle-aged married woman who frequently shops online, actively engages on social media, and enjoys mobile gaming on her tablet. She has a preference for eco-friendly brands, enjoys mystery movies, and values sustainability and wellness. She is situated in a cozy apartment in downtown Toronto, Canada.”

(277) This prompt provides a comprehensive view of the target audience, painting a vivid picture that encompasses demographic, behavioral, psychographic, and geographic attributes.

(278) However, the granularity of the prompt will depend on the specificity of the data the user has

provided. The more detailed the data, the more tailored the prompt can be, allowing for a more targeted image generation.

(279) There are generic guidelines. Use the user-provided information and your expertise as an AI Assistant in generating good prompts for the generation of images with Dall-e or stable diffusion to help generate the same.

(280) The user can provide additional information with user instructions for the CGS to generate images. This information may be derived from information about products, audiences, etc., configured by the user on DGS. In the example below, the information contains the product with a description of the customer, an audience definition with a description, and some relevant prompt examples.

(281) You are an AI Assistant who will help generate good image-generation prompts.

(282) I sell the following products: 1. Barista-101—Espresso machine for home. 2. OffEspro-101—Espresso machine for office. 3. KidsChoc-101—Hot chocolate vending machine for kids.

(283) Following are the descriptions of my audiences: 1. Goldstar—Home users from North America who use Barista-101 machines. 2. KidChamp—Kids from California who use KidsChoc-101 3. OffPro—Office users from New York who use OffEspro-101

(284) Incorporate some key attributes or adjectives of the product in the prompt generated.

(285) Incorporate some key attributes or adjectives of the audience in the prompt generated.

(286) Some prompt examples below. Don't use the same as is but use them as inspiration:

(287) Photo of [XM-Coffee-machine] on a wooden table, bakery in the background, streets of London elegant, perfect composition, perfect composition, beautiful detailed intricate, insanely detailed, 8k artistic photography, photorealistic, soft natural cinematic lighting.

(288) In one example, the user prompt is a single-line goal, “I want to promote a home coffee machine for Halloween.”

(289) The prompt generated CGS is: “A photo of the Barista-101 espresso machine, adorned with Halloween-themed decorations like cobwebs and mini pumpkins. The machine is set in a cozy North American home kitchen, with a steaming cup of espresso on the side. The atmosphere is warm and inviting, with soft lighting and autumnal colors, perfect for a Facebook post to engage with the Goldstar audience.”

(290) In one example, the CGS generated four images of the espresso machine based on this prompt.

(291) In another example, the user prompt is “I want to promote office coffee machine for Christmas holidays targeting Instagram channel.”

(292) The prompt generated CGS is: “A photo of the OffEspro-101 espresso machine, decorated with Christmas ornaments and twinkling fairy lights. The machine is placed in a bustling New York office setting, with a steaming cup of espresso and a plate of Christmas cookies on the side. The atmosphere is festive and energetic, with a beautifully decorated Christmas tree in the background. The image is vibrant and aesthetically pleasing, perfect for an Instagram post to engage with the OffPro audience.”

(293) In one example, the CGS generated four images of the espresso machine with holiday backgrounds based on this prompt.

(294) In another example, the CGS generates an email for a marketing campaign. An example prompt template is as follows:

(295) You are an AI Assistant that will help generate email for my email marketing campaign.

(296) Creating an email for a marketing campaign requires gathering specific information from the user to ensure that the campaign is well-targeted and effective. Here's a structure for the brief, along with the inputs you should request from the user:

(297) Below is an example structured template for a marketing email promoting a product for Halloween. Keep in mind that the specifics can be tailored to suit the company's brand voice, product features, and campaign goals.

(298) ****Subject Line:**** Unlock Spooky Savings This Halloween with [Product Name]!

(299) ****Preheader Text:**** Dive into the Halloween spirit! Enjoy exclusive deals on [Product Name] for a limited time.

(300) ****[Email Header Image: A visually appealing and themed image showcasing the product.]****

(301) ****Hi [First Name],****

(302) ****[Introduction: Capture their attention and set the tone]****

(303) Get ready to make this Halloween spook-tacular with [Product Name]! Whether you're planning a haunted house extravaganza or a cozy night in, [Product Name] is the perfect companion for your festivities.

(304) ****[Product Showcase: Highlight the product's key features and benefits]**** ****Spooky Feature #1:**** [Brief description] ****Spooky Feature #2:**** [Brief description] ****Spooky Feature #3:**** [Brief description]

[Product Image]

(305) With [Product Name], you're guaranteed an unforgettable Halloween experience, packed with [mention any specific benefits or features].

(306) *****: [Special Offer: Present the Halloween promotion or special deal]**

(307) For a limited time only, enjoy a frighteningly good discount of [X] % when you purchase [Product Name] using the code SPOOKYSAVINGS at checkout!

(308) ****[Testimonials/Reviews: Share social proof]****

(309) But don't just take our word for it! Here's what our happy customers have to say about [Product Name]:

(310) ****“I can't imagine Halloween without [Product Name]. It totally transformed our celebration!”—[Customer Name]****

(311) ****[Call-to-Action: Encourage them to take action]****

(312) Ready to elevate your Halloween? Click the button below to grab your [Product Name] at this exclusive price!

(313) ****[CTA Button: “Shop Now” or “Get My Deal”]****

(314) ****[Additional Information: Provide any necessary details about the promotion]****

(315) This haunting deal won't last long-the clock strikes midnight on [Date], and the savings will vanish!

(316) ****:** [Footer: Include contact information and legal compliance]**

(317) Stay spooky,

(318) [Your Brand Name]

(319) [Your Company's Physical Address]

(320) [Unsubscribe Link]

(321) ****[Optional: Social Media Links]****

(322) Connect with us:

(323) [Social Media Icons/Links]

(324) This structure aims to create a sense of excitement, highlight the product's benefits, provide social proof, and encourage immediate action, all while maintaining a festive Halloween theme. Ensure that the visuals, tone, and content align with your brand identity and resonate with your target audience.

(325) Below is an example of user-provided information for generating the prompt:

(326) You are an AI Assistant who will help generate good image-generation prompts.

(327) I sell the following products: 1. Barista-101—Espresso machine for home. 2. OffEspro-101—Espresso machine for office. 3. KidsChoc-101—Hot chocolate vending machine for kids.

(328) Following are the descriptions of my audiences: 1. Goldstar—Home users from North America who use Barista-101 machines. 2. KidChamp—Kids from California who use KidsChoc-101 3. OffPro—Office users from New York who use OffEspro-101

(329) Incorporate some key attributes or adjectives of the product in the prompt generated.

(330) Incorporate some key attributes or adjectives of the audience in the prompt generated.

(331) Use the following brandkit information where appropriate

(332) Normal font: Calibri

(333) Header font: Calibri

(334) Logo small: <https://examplecompany.com/logo.png>

(335) Brand primary color pallet: #77DD7

(336) Use the following email HTML template structure to fill up the relevant content. Make sure the header, footer, logo, and style are maintained as part of the generation

```
(337) TABLE-US-00001  <html>      <head>          <img src
https://examplecompany.com/logo.png>      </head>      <body>          <div>      {{ fill this with
the content of max 512 length on product description targeting the audience}}      </div>
<img src={{ AI Assistant fill the link to the image required in format -- https://cdn.typeface.ai?
product='productname' & audience ='audience description' & prompt='a simple prompt to
generate image with dale-3' & width=512&height=512 }}/>      <div style='background-
color:green'>          {{ fill up call to action over here}}      </div>      </body>      <footer>
Copyright www.examplecompany.com      </footer>      </html>
```

(338) Example output 1:

```
(339) TABLE-US-00002  <html>      <head>                </head>      <body>          <div>          Hi [First
Name],          <p>
```

(340) Transform your kitchen into a professional cafe this Halloween with Barista-101! Specially designed for coffee enthusiasts in North America, this espresso machine brings quality, efficiency, and the spirit of a barista right to your home.

```
(341) TABLE-US-00003          </p>          </div>                    <div style='background-color:green'>
<p>          Ready to brew like a pro? Click below to unlock your exclusive
Halloween discount on Barista-101!          </p>          <button style='color:white'>Get My
Deal</button>          </div>          </body>          <footer>          Copyright
www.coffemachineco.com          </footer>          </html>  Example output 2 for kids:  <html>
```

```
<head>                    </head>          <body>
<div>          Hi [First Name],          <p>
```

(342) Make every coffee break a treat this Halloween with OffEspro-101! Designed for the fast-paced environment of New York offices, this espresso machine ensures a quick, smooth, and delightful coffee experience every time.

```
(343) TABLE-US-00004          </p>          </div>          
<div style='background-color:green'>          <p>          Upgrade your office coffee
experience today! Click below to snag your OffEspro- 101 at an unbeatable Halloween price!
</p>          <button style='color:white'>Get My Deal</button>          </div>
</body>          <footer>          Copyright www.examplecompany.com          </footer>          </html>
```

(344) Here is another example of a prompt template for generating a post for job recruiters to be posted on a social network or a jobs website:

(345) TABLE-US-00005 POST_HR = BasePromptBlueprint(task_description="Write a job

(351) Once a feed is established, FG can apply the parameters of the feed to different app-level patterns, such as email journeys, orchestrations, social media, and paid media. If there is a marketing job that needs to be done, the FG can help a growth marketer by creating advertisements. FG works with the user to create a customized solution that utilizes feeds as a core and optimizes the experience level to fit the user's specific workflows. For example, if the user needs an integration into the email system, FG can facilitate that.

(352) There are two ways in which the creation of new content is triggered: when a request for new content is received at operation **2402** and when the creation of new content is scheduled at operation **2404**, e.g., generation of content every night while the system is less busy, so the user can access and select new content generated automatically by the FG.

(353) From operation **2404**, the method **2400** flows to operation **2406** to identify the target of the communication that will use the generated content, e.g., selecting the project and audience for the communication. The process may be repeated multiple times for each of the different projects and active audiences.

(354) From operation **2406** and operation **2402**, the method **2400** flows to operation **2408** to identify the attributes of the recipient (e.g., demographics, channel, brand parameters, user curation history).

(355) From operation **2408**, method **2400** flows to operation **2410**, which creates a prompt for the GAI tool.

(356) From operation **2410**, the method **2400** flows to operation **2412**, where the content is generated using the selected GAI tool

(357) From operation **2412**, the method **2400** flows to operation **2414** to store the generated content in memory, e.g., a database.

(358) From operation **2414**, the method **2400** flows to operation **2416**, where the new content is presented to the user, e.g., in the canvas tool of FIG. 1. After the new content is presented, the content creation may be repeated by reentering the method **2400** at operation **2402** or operation **2404**.

(359) From operation **2416**, the method **2400** flows to operation **2418**, where a request is received to send the content item in a communication.

(360) From operation **2418**, the method **2400** flows to operation **2420**, where the content is transmitted for presentation to an end-user.

(361) Further, the FG can leverage the campaign journey orchestration that has been defined in a third-party system. FG gathers data from these systems, compiles it, and then creates a personalized content feed as indicated in the method **2400**.

(362) FIG. 25 is a flowchart of a method **2500** for generating personalized content according to some examples. While the various operations in this flowchart are presented and described sequentially, one of ordinary skill will appreciate that some or all of the operations may be executed in a different order, be combined or omitted, or be executed in parallel.

(363) To track customer communications, the user may select to perform A/B testing with FG. The A/B testing involves testing two different versions of an item (e.g., an Instagram post) and comparing their relative performance, e.g., which item produced better results, such as a higher clickthrough rate. However, with the ease of creating new content, the user can perform multiple rounds of A/B testing during a campaign in order to optimize the content performance continuously. By doing so, the user can create a feedback loop to allow the ongoing improvement of the campaign's effectiveness.

(364) The FG is designed to prompt users for content and generate personalized content based on data feeds and previous campaigns. As the content creation process becomes more streamlined, the focus may shift from creation to curation, allowing users to sift through content to find what they like best without having even to request the generation of new content.

(365) How does the user improve the content used in campaigns? There are two parts: the FG

creates and transmits personalized content, and the experience gained from the performance of past communications is used to curate the feed content.

(366) There are two parts to finding what the user prefers: first, there is an engine that creates personalized content, and second, there is an experience-level innovation to curate a feed of content. With the ability to generate a large number of content items (e.g., hundreds of items), FG can create batches of bulk creations to create content in the background, even while the user is not logged in. Every day, the user can see new ideas and decide which ones to select. Further, based on the user choices, the user can request more options and set up performance testing. This is how the experiences and content evolve and improve over time.

(367) Operation **2502** is for transmitting one or more content items to users.

(368) From operation **2502**, the method **2500** flows to operation **2504** to determine the attributes of the content items (e.g., the content of an image, text sent, the layout of the communication, frequency of communications, and the audience).

(369) From operation **2504**, the method **2500** flows to operation **2506** to measure the performance of the content items based on the content attributes. As discussed above, A/B testing can be used to test the performance based on the selected predetermined performance parameters (e.g., clickthroughs to a website included in the communication, amount of time spent on the communication, number of visits to the website, number of sales, total amount of sales, number of shares of the communication with other users, number of responses sent to the communication).

(370) From operation **2506**, the method **2500** flows to operation **2508** to select the attribute values for new content to be generated based on the measured performance. In some examples, a machine learning (ML) model may be used to select the attribute values that perform best given the item promoted and the target audience. More details regarding the ML model are provided below with reference to FIG. 27.

(371) In other examples, rules may be used to select attributes based on past performance; however, given the large number of options in the system, it may be difficult to create a large number of rules for selecting content, and that is why the ML model works best in complex environments.

However, rules may also be valuable in smaller environments with less complexity (e.g., a small number of products, a small number of campaigns, and a small number of customer communications).

(372) From operation **2508**, the method **2500** flows to operation **2510** to generate new content items based on the selected attributes (e.g., ocean background, friendly-tone text).

(373) From operation **2510**, the method **2500** flows to operation **2512** to present the generated new content to the user.

(374) From operation **2512**, the method **2500** flows to operation **2514** to receive a request to send selected content to customers.

(375) From operation **2514**, the method **2500** flows to operation **2502** to transmit the one or more content items.

(376) FIG. 26 is a flowchart of a method **2600** for details on generating the personalized content according to some examples. While the various operations in this flowchart are presented and described sequentially, one of ordinary skill will appreciate that some or all of the operations may be executed in a different order, be combined or omitted, or be executed in parallel.

(377) Operation **2602** is for accessing campaign information and performance data. As described above with reference to FIG. 23, there are two ways to improve the outputs: by improving the GAI prompt **2310** or by fine-tuning **2314** the GAI model with user data. For example, after running several campaigns, performance data shows that images with water backgrounds perform better. Afterward, the GAI prompt **2310** will include text requesting a water background: e.g., “. . . create an image with water background . . . and the following attributes: . . .”. Further, the GAI model **2312** will be improved with the performance data from previous campaigns.

(378) From operation **2602**, the method **2600** flows to operation **2604** to select the target recipient

of the communication, e.g., a user or an audience.

(379) From operation **2604**, the method **2600** flows to operation **2606** to determine the recipient's attributes. To create personalized content, the user attributes are determined. There may be multiple sources of attributes about the user. In some examples, the recipient's attributes include data from one or more of campaign, audience, author, performance data of the content previously sent, geography, channel, demographics, brand, usurped duration history, etc.

(380) The campaign data refers to information about past campaigns, such as channel used, audiences, content sent, scheduling of communications, etc. The author refers to the person who is sending the communication, e.g., the CEO of the company is the sender of the communication, a technical lead, or a support person.

(381) The performance data refers to the performance of previous communications, such as the number of users who responded to the communication. Geography refers to the location of the recipient, as the local messaging styles may differ from country to country, from large cities to small cities, etc.

(382) The channel refers to the media that will be used for the communication, such as email, texting, social media posts, advertisements, etc. The demographics refer to information about the user, such as age, gender, city of residence, etc.

(383) The brand refers to the requirements set up by the user for the brand, such as professional content, casual content, the brand's values, etc. Further, the user curation history refers to the selections made by users when content was previously suggested to them, e.g., what content was discarded and what content was used for communications.

(384) The FG will determine what attributes are important for the communication, e.g., when the tone of the content is friendly and the images used are related to water, the click-through rate for this particular audience segment increases. With the use of AI, it is possible to analyze the images and identify their contents, such as a blue chair, a black dress, a pool, or a sky. This is a valuable insight generated by tracking performance instead of having to rely on user feedback.

(385) From operation **2606**, the method **2600** flows to operation **2608** to generate personalized content for the user.

(386) FIG. **27** illustrates the training and use of a machine-learning model **2716**, according to some example examples. In some examples, machine learning (ML) models **2716** are utilized to select attributes for generating content.

(387) Machine Learning (ML) is an application that provides computer systems the ability to perform tasks without explicitly being programmed by making inferences based on patterns found in the analysis of data. Machine learning explores the study and construction of algorithms, also referred to herein as tools, that may learn from existing data and make predictions about new data. Such machine-learning algorithms operate by building an ML model **2716** from training data **2712** in order to make data-driven predictions or decisions expressed as outputs or assessments **2720**. Although examples are presented with respect to a few machine-learning tools, the principles presented herein may be applied to other machine-learning tools.

(388) The performance data feeds into the generative AI models to improve the content creation itself. As often is the case with GAI, the GAI models improve over time as more people use them. Therefore, every campaign is a learning opportunity, and the subsequent campaign will start with better content than in the previous campaign. In some examples, the ML model **2716** provides a score of attributes that can be used to generate content. The attributes with the highest scores are then used to create the prompts for the GAI tools.

(389) There are two common modes for ML: supervised ML and unsupervised ML. Supervised ML uses prior knowledge (e.g., examples that correlate inputs to outputs or outcomes) to learn the relationships between the inputs and the outputs. The goal of supervised ML is to learn a function that, given some training data, best approximates the relationship between the training inputs and outputs so that the ML model can implement the same relationships when given inputs to generate

the corresponding outputs. Unsupervised ML is the training of an ML algorithm, using information that is neither classified nor labeled and allowing the algorithm to act on that information without guidance. Unsupervised ML is useful in exploratory analysis because it can automatically identify structure in data.

(390) Typical tasks for supervised ML are classification problems and regression problems. Classification problems, also referred to as categorization problems, aim to classify items into one of several category values (for example, is this object an apple or an orange?). Regression algorithms aim to quantify some items (for example, by providing a score to the value of some input). Some examples of commonly used supervised ML algorithms are Logistic Regression (LR), Naive-Bayes, Random Forest (RF), neural networks (NN), deep neural networks (DNN), matrix factorization, and Support Vector Machines (SVM).

(391) Some typical tasks for unsupervised ML include clustering, representation learning, and density estimation. Some examples of commonly used unsupervised ML algorithms are K-means clustering, principal component analysis, and autoencoders.

(392) The training data **2712** comprises examples of values for the features **2702**. In some examples, the training data comprises labeled data with examples of values for the features **2702** and labels indicating the outcome, such as the user clicked on the content, the user made a purchase, the user asked for additional information, etc. The machine-learning algorithms utilize the training data **2712** to find correlations among identified features **2702** that affect the outcome. A feature **2702** is an individual measurable property of a phenomenon being observed. The concept of a feature is related to that of an explanatory variable used in statistical techniques such as linear regression. Choosing informative, discriminating, and independent features is essential for the effective operation of ML in pattern recognition, classification, and regression. Features may be of different types, such as numeric, strings, categorical, and graph. A categorical feature is a feature that may be assigned a value from a plurality of predetermined possible values (e.g., this animal is a dog, a cat, or a bird).

(393) In one example, the features **2702** may be of different types. They may include data from one or more of campaign **2703**, audience **2704**, author **2705**, content performance **2706**, channel **2707**, demographics **2708**, brand **2709**, user history **2710** (e.g., content selected by the user), etc. The layout is another feature that may be considered, e.g., for communications with the same content, which layout performed better: the image on top, the image on the left side of the text, the image on the center of the text, the image on the right side of the text, the image below the text. When a certain layout performs the best for a certain audience, this layout will be given preference in future communications to this audience.

(394) Features related to the content may also be included. The content created can take on various forms, such as text, images, or videos. Each form has its own unique attributes. For text, for instance, one consideration is the length of the text. Is it better to use short-form or long-form text? This will depend on the type of content and the intended audience. Additionally, the tone used is another feature because the tone will vary depending on the audience. For example, different tones would be used when communicating with a teenager versus when communicating with a professional.

(395) For video, multiple attributes define a video, such as whether animations work better than still images, what the length of the video is, whether there is sound with the video, etc.

(396) Regarding author **2705** attributes, the communication will account for who the author or sender behind the message is. For example, if a CEO communicates, people may expect the CEO to be inspirational, versus a communication from a technical lead will have more technical content.

(397) The characteristics of the brand **2709** relate to the guidelines used for communications associated with the brand, such as the set of brand values and tones that the marketers are consistently using or avoiding. Thus, it is important to ensure that the brand's voice and tone are being reflected in the best possible way so they are being reflected accurately.

(398) Typically, there is a process of personalizing a brand. As a brand, the marketer may choose to use warm tones instead of cool ones or use vector graphics for the background instead of real photography. All of these preferences can be learned based on measuring the performance, e.g., what works well and what does not work well.

(399) The content performance **2706** of the content previously presented is also a valuable feature for selecting content attributes. For example, what image resolution works best, does landscape perform better than portrait orientation, etc. Additionally, what is in the image (e.g., the background of the image, whether it is a city, forest, or ocean), and how does it affect performance for each audience?

(400) In some examples, AI tools for computer vision are utilized to identify the contents of the images, and these tools are sometimes referred to as visual Q&A systems. For example, what is the color of the T-shirt shown in the image? Is there a pattern on the T-shirt? etc. The FG may utilize the tool to determine what is in the image and then use the content identified for performance tracking that may be used in the training data **2712**.

(401) During training **2714**, the ML program, also referred to as ML algorithm or ML tool, analyzes the training data **2712** based on identified features **2702** and configuration parameters defined for the training. The result of the training **2714** is the ML model **2716**, which is capable of taking inputs to produce assessments.

(402) Training an ML algorithm involves analyzing large amounts of data (e.g., from several gigabytes to a terabyte or more) in order to find data correlations. The ML algorithms utilize the training data **2712** to find correlations among the identified features **2702** that affect the outcome or assessment **2720**. In some examples, the training data **2712** includes labeled data, which is known data for one or more identified features **2702** and one or more outcomes. In some examples, the training data **2712** includes content items previously created (including the attributes associated with the content), and the outcomes include performance metrics of the content items (e.g., items were selected for transmittal or not, number of clickthroughs on the communications sent, number of sales, number of visits to the website referenced in the communication, etc.).

(403) The ML algorithms usually explore many possible functions and parameters before finding what the ML algorithms identify to be the best correlations within the data; therefore, training may make use of large amounts of computing resources and time.

(404) When the ML model **2716** is used to perform an assessment, new data **2718** is provided as input to the ML model **2716**, and the ML model **2716** generates the assessment **2720** as output. In some examples, the input for the ML model **2716** includes information about an attribute, and that assessment **2720** provides a score (e.g., a number from 1 to 100, a real number between 0 and 1) regarding the relevance of the attribute for generating content.

(405) In other examples, the input to the ML model **2716** includes a list of attributes (e.g., represented within a vector), and the output provides scores for the list of attributes. In another example, the output includes a list of a predetermined number of the best attributes for generating content. The list of attributes of the input may include the audience that is targeted for the communication.

(406) In some examples, results obtained by the model **2716** during operation (e.g., assessment **2720** produced by the model in response to inputs) are used to improve the training data **2712**, which is then used to generate a newer version of the model. Thus, a feedback loop is formed to use the results obtained by the model to improve the model.

(407) FIG. **28** is a flowchart of a method **2800** for proactively generating personalized content according to some examples. While the various operations in this flowchart are presented and described sequentially, one of ordinary skill will appreciate that some or all of the operations may be executed in a different order, be combined or omitted, or be executed in parallel.

(408) Operation **2802** is for receiving campaign parameters to create a campaign for a product or service. The campaign parameters comprise a goal, one or more channels, and an audience.

(409) From operation **2802**, the method **2800** flows to operation **2804** for creating the campaign based on the campaign parameters.

(410) From operation **2804**, the method **2800** flows to operation **2806** to proactively generate, using one or more generative artificial intelligence (GAI) tools, items based on the campaign parameters, the items being multimodal, wherein the created items are included in a feed.

(411) From operation **2806**, the method **2800** flows to operation **2808** for providing a user interface (UI) with an option to select one or more items of the feed for inclusion in a journey associated with the campaign, the journey comprising a plurality of communications to a user based on the campaign parameters.

(412) From operation **2808**, the method **2800** flows to operation **2810** for receiving a selection of selected feed items for the journey.

(413) From operation **2810**, the method **2800** flows to operation **2812** for transmitting the selected feed items to the audience of the campaign based on the journey.

(414) In some examples, the method **2800** further comprises providing an option in the UI to generate additional items for the feed based, the UI comprising a field for entering user instructions to generate the additional items.

(415) In some examples, proactively generating the items further comprises, for each item to be generated, selecting a GAI tool based on a type of item to be generated; creating a prompt to generate the item; and providing the prompt as input to the GAI tool that generates the item as output.

(416) In some examples, the method **2800** further comprises providing an option in the UI to access an existing plan created with a third-party tool and obtaining the campaign parameters using the third-party tool.

(417) In some examples, the items are generated based on brand parameters for the product or service.

(418) In some examples, the multimodal items are configured to include text items, image items, sound items, and video items.

(419) In some examples, the campaign parameters further comprise one or more channels for the distribution of the content.

(420) In some examples, creating the campaign further comprises identifying messages for the campaign, determining channels for communication, and creating a timeline for execution.

(421) In some examples, image items generated for the campaign comprise images generated based on a prompt entered by the user and images proactively generated.

(422) In some examples, the method **2800** further comprises providing an audience UI that presents audience information of the feed in a table, wherein the audience information is organized by having an audience type in each row and communications for the audience type in the row.

(423) Another general aspect is for a system that includes a memory comprising instructions and one or more computer processors. The instructions, when executed by the one or more computer processors, cause the one or more computer processors to perform operations comprising: receiving campaign parameters to create a campaign for a product or service, the campaign parameters comprising a goal, one or more channels, and an audience; creating the campaign based on the campaign parameters; proactively generating, using one or more generative artificial intelligence (GAI) tools, items based on the campaign parameters, the items being multimodal, wherein the created items are included in a feed; providing a user interface (UI) with an option to select one or more items of the feed for inclusion in a journey associated with the campaign, the journey comprising a plurality of communications to a user based on the campaign parameters; receiving a selection of selected feed items for the journey; and transmitting the selected feed items to the audience of the campaign based on the journey.

(424) In yet another general aspect, a tangible machine-readable storage medium (e.g., a non-transitory storage medium) includes instructions that, when executed by a machine, cause the

machine to perform operations comprising: receiving campaign parameters to create a campaign for a product or service, the campaign parameters comprising a goal, one or more channels, and an audience; creating the campaign based on the campaign parameters; proactively generating, using one or more generative artificial intelligence (GAI) tools, items based on the campaign parameters, the items being multimodal, wherein the created items are included in a feed; providing a user interface (UI) with an option to select one or more items of the feed for inclusion in a journey associated with the campaign, the journey comprising a plurality of communications to a user based on the campaign parameters; receiving a selection of selected feed items for the journey; and transmitting the selected feed items to the audience of the campaign based on the journey.

(425) FIG. **29** is a flowchart of a method **2900** for proactively generating content based on the tracked performance of previous content, according to some examples. While the various operations in this flowchart are presented and described sequentially, one of ordinary skill will appreciate that some or all of the operations may be executed in a different order, be combined or omitted, or be executed in parallel.

(426) Operation **2902** is for transmitting a plurality of first items for presentation to users, the first items being multimodal, each first item having values for one or more attributes associated with the first item.

(427) From operation **2902**, method **2900** flows to operation **2904** to track the performance of the transmitted plurality of first items.

(428) From operation **2904**, the method **2900** flows to operation **2906** for selecting values of one or more attributes based on the tracked performance of the plurality of first items.

(429) From operation **2906**, the method **2900** flows to operation **2908** for proactively generating, using one or more generative artificial intelligence (GAI) tools, a plurality of second items based on the selected values of the one or more attributes.

(430) From operation **2908**, the method **2900** flows to operation **2910** for providing a user interface (UI) with an option to select from the plurality of second items.

(431) From operation **2910**, the method **2900** flows to operation **2912** for receiving in the UI a selection of selected second items for transmittal.

(432) From operation **2912**, the method **2900** flows to operation **2914** for transmitting the selected second items to one or more users.

(433) Given the disclosure above, various examples are set forth below. It should be noted that one or more features of an example, taken in isolation or combination, should be considered within the disclosure of this application.

(434) In some examples, the method **2900** further comprises training a machine-learning model with training data comprising performance information of items, the machine-learning model generating relevance scores of attributes for generating items.

(435) In some examples, the machine-learning model receives as input information about an attribute and produces an output that is a score regarding a relevance of the attribute for generating content.

(436) In some examples, the one or more attributes comprise one or more of campaign, audience, author, content performance, channel, demographics, or brand.

(437) In some examples, proactively generating a plurality of second items further comprises, for each second item to be generated: selecting a GAI tool based on a type of item to be generated; creating a prompt to generate the second item; and providing the prompt as input to the GAI tool that generates the second item as output.

(438) In some examples, the method **2900** further comprises receiving values of user attributes for the recipient of a communication; and selecting one or more items from the plurality of second items for presentation on a user interface (UI) based on the user attributes and the tracked performance of the plurality of first items.

(439) In some examples, the method **2900** further comprises receiving values of audience attributes

for an audience of a communication; and selecting one or more items from the plurality of second items for presentation on a user interface (UI) based on the audience attributes and the tracked performance of the plurality of first items.

(440) In some examples, the method **2900** further comprises providing a UI for presenting performance information for the plurality of first items, the UI providing a table with content for each type of audience, the content being ranked according to performance.

(441) In some examples, the method **2900** further comprises providing a UI for adding content to a feed of items, the content being selected from the plurality of second items.

(442) In some examples, the method **2900** further comprises providing a UI for presenting information about the tracked performance, the information comprising insights based on the tracked performance for placement of items.

(443) Another general aspect is for a system that includes a memory comprising instructions and one or more computer processors. The instructions, when executed by the one or more computer processors, cause the one or more computer processors to perform operations comprising: transmitting a plurality of first items for presentation to users, the first items being multimodal, each first item having values for one or more attributes associated with the first item; tracking performance of the transmitted plurality of first items; selecting values of one or more attributes based on the tracked performance of the plurality of first items; proactively generating, using one or more generative artificial intelligence (GAI) tools, a plurality of second items based on the selected values of the one or more attributes; providing a user interface (UI) with an option to select from the plurality of second items; receiving in the UI a selection of selected second items for transmittal; and transmitting the selected second items to one or more users.

(444) In yet another general aspect, a tangible machine-readable storage medium (e.g., a non-transitory storage medium) includes instructions that, when executed by a machine, cause the machine to perform operations comprising: transmitting a plurality of first items for presentation to users, the first items being multimodal, each first item having values for one or more attributes associated with the first item; tracking performance of the transmitted plurality of first items; selecting values of one or more attributes based on the tracked performance of the plurality of first items; proactively generating, using one or more generative artificial intelligence (GAI) tools, a plurality of second items based on the selected values of the one or more attributes; providing a user interface (UI) with an option to select from the plurality of second items; receiving in the UI a selection of selected second items for transmittal; and transmitting the selected second items to one or more users.

(445) FIG. **30** is a flowchart of a method **3000** for generating personalized content, according to some examples. While the various operations in this flowchart are presented and described sequentially, one of ordinary skill will appreciate that some or all of the operations may be executed in a different order, be combined or omitted, or be executed in parallel.

(446) Operation **3002** is for identifying audience parameter values, the audience parameter values indicating which users are members of an audience.

(447) From operation **3002**, the method **3000** flows to operation **3004** for determining attribute values for a plurality of attributes used to generate items based on the audience parameter values.

(448) From operation **3004**, the method **3000** flows to operation **3006** for generating a plurality of items for the audience based on the attribute values. Operation **3006** comprises operations **3008**, **3010**, and **3012**.

(449) Operation **3008** is for creating a prompt based on a type of the item, the attribute values, and the audience parameter values.

(450) From operation **3008**, the method **3000** flows to operation **3010** for selecting a generative artificial intelligence (GAI) tool to generate the item based on the type of item.

(451) From operation **3010**, the method **3000** flows to operation **3012** for providing the created prompt to the selected GAI tool to obtain the item.

(452) From operation **3012**, the method **3000** flows to operation **3014** for causing the presentation of the generated plurality of items in a user interface (UI).

(453) From operation **3014**, the method **3000** flows to operation **3016** for receiving on the UI a selection of one of the items from the generated plurality of items.

(454) From operation **3016**, the method **3000** flows to operation **3018** for transmitting the selected item to one or more members of the audience.

(455) Given the disclosure above, various examples are set forth below. It should be noted that one or more features of an example, taken in isolation or combination, should be considered within the disclosure of this application.

(456) In some examples, the method **3000** further comprises providing a first user interface (UI) displaying information regarding different audiences, audience preferences, and content designed for each audience, the first UI providing an option to select and filter information by audience.

(457) In some examples, the first UI provides cards showing documents generated for each communication channel.

(458) In some examples, the method **3000** further comprises providing a second UI showing a performance of content presented to users and a feed of items available for future communications to the audience.

(459) In some examples, the performance is a measurement of a response of users to communications, and the performance is based on one or more of a number of views, a percentage of users that selected a link for additional information, a percentage of users that bought a product or service associated with a transmitted item.

(460) In some examples, generating each item further comprises utilizing past performance data to generate the item.

(461) In some examples, determining attribute values further comprises utilizing an image-recognition model to determine features of images sent to users, determining a relevance score for the features based on the tracked performance of the images sent to users, and selecting the attribute values based on the relevance score of the features.

(462) In some examples, the image-recognition model is trained using a dataset of images with known performance metrics, the image-recognition model learning correlations between visual elements and item performance, wherein the image-recognition model predicts a performance of new images based on visual features of the new images.

(463) In some examples, the method **3000** further comprises providing a second UI showing audience information on a feed, the second UI showing content organized by audience for a plurality of audiences, the second UI presenting the audience information in a table organized by calendar date, with each column for a different day and each row showing communications planned or delivered for each day for each audience.

(464) In some examples, the method **3000** further comprises generating a feed for the audience, wherein the items generated for the feed are deliverable via a plurality of channels that comprise publishing to social media, connecting to email automation systems, providing audience-level personalization on a webpage, and distributing through customer journey orchestrators.

(465) Another general aspect is for a system that includes a memory comprising instructions and one or more computer processors. The instructions, when executed by the one or more computer processors, cause the one or more computer processors to perform operations comprising: identifying audience parameter values, the audience parameter values indicating which users are members of an audience; determining attribute values for a plurality of attributes used to generate items based on the audience parameter values; generating a plurality of items for the audience based on the attribute values, wherein generating each item from the plurality of items comprises: creating a prompt based on a type of the item, the attribute values, and the audience parameter values; selecting a generative artificial intelligence (GAI) tool to generate the item based on the type of item; and providing the created prompt to the selected GAI tool to obtain the item; causing

presentation of the generated plurality of items in a user interface (UI); receiving on the UI a selection of one of the items from the generated plurality of items; and transmitting the selected item to one or more members of the audience.

(466) In yet another general aspect, a tangible machine-readable storage medium (e.g., a non-transitory storage medium) includes instructions that, when executed by a machine, cause the machine to perform operations comprising: identifying audience parameter values, the audience parameter values indicating which users are members of an audience; determining attribute values for a plurality of attributes used to generate items based on the audience parameter values; generating a plurality of items for the audience based on the attribute values, wherein generating each item from the plurality of items comprises: creating a prompt based on a type of the item, the attribute values, and the audience parameter values; selecting a generative artificial intelligence (GAI) tool to generate the item based on the type of item; and providing the created prompt to the selected GAI tool to obtain the item; causing presentation of the generated plurality of items in a user interface (UI); receiving on the UI a selection of one of the items from the generated plurality of items; and transmitting the selected item to one or more members of the audience.

(467) FIG. 31 is a block diagram illustrating an example of a machine **3100** upon or by which one or more example process embodiments described herein may be implemented or controlled. In alternative embodiments, the machine **3100** may operate as a standalone device or be connected (e.g., networked) to other machines. In a networked deployment, the machine **3100** may operate in the capacity of a server machine, a client machine, or both in server-client network environments. For example, the machine **3100** may act as a peer machine in a peer-to-peer (P2P) (or other distributed) network environment. Further, while only a single machine **3100** is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein, such as via cloud computing, software as a service (SaaS), or other computer cluster configurations.

(468) Examples, as described herein, may include, or may operate by, logic, various components, or mechanisms. Circuitry is a collection of circuits implemented in tangible entities, including hardware (e.g., simple circuits, gates, logic). Circuitry membership may be flexible over time and underlying hardware variability. Circuitries include members that may, alone or in combination, perform specified operations when operating. In an example, the hardware of the circuitry may be immutably designed to carry out a specific operation (e.g., hardwired). In an example, the hardware of the circuitry may include variably connected physical components (e.g., execution units, transistors, simple circuits), including a computer-readable medium physically modified (e.g., magnetically, electrically, by moveable placement of invariant massed particles) to encode instructions of the specific operation. In connecting the physical components, the underlying electrical properties of a hardware constituent are changed (for example, from an insulator to a conductor or vice versa). The instructions enable embedded hardware (e.g., the execution units or a loading mechanism) to create members of the circuitry in hardware via the variable connections to carry out portions of the specific operation when in operation. Accordingly, the computer-readable medium is communicatively coupled to the other circuitry components when the device operates. In an example, any of the physical components may be used in more than one member of more than one circuitry. For example, under operation, execution units may be used in a first circuit of a first circuitry at one point in time and reused by a second circuit in the first circuitry or by a third circuit in a second circuitry at a different time.

(469) The machine **3100** (e.g., computer system) may include a hardware processor **3102** (e.g., a central processing unit (CPU), a hardware processor core, or any combination thereof), a graphics processing unit (GPU **3103**), a main memory **3104**, and a static memory **3106**, some or all of which may communicate with each other via an interlink **3108** (e.g., bus). The machine **3100** may further include a display device **3110**, an alphanumeric input device **3112** (e.g., a keyboard), and a user

interface (UI) navigation device **3114** (e.g., a mouse). In an example, the display device **3110**, alphanumeric input device **3112**, and UI navigation device **3114** may be a touch screen display. The machine **3100** may additionally include a mass storage device **3116** (e.g., drive unit), a signal generation device **3118** (e.g., a speaker), a network interface device **3120**, and one or more sensors **3121**, such as a Global Positioning System (GPS) sensor, compass, accelerometer, or another sensor. The machine **3100** may include an output controller **3128**, such as a serial (e.g., universal serial bus (USB)), parallel, or other wired or wireless (e.g., infrared (IR), near field communication (NFC)) connection to communicate with or control one or more peripheral devices (e.g., a printer, card reader).

(470) The processor **3102** refers to any one or more circuits or virtual circuits (e.g., a physical circuit emulated by logic executing on an actual processor) that manipulates data values according to control signals (e.g., commands, opcodes, machine code, control words, macroinstructions, etc.) and which produces corresponding output signals that are applied to operate a machine. A processor **3102** may, for example, include at least one of a Central Processing Unit (CPU), a Reduced Instruction Set Computing (RISC) Processor, a Complex Instruction Set Computing (CISC) Processor, a Graphics Processing Unit (GPU), a Digital Signal Processor (DSP), a Tensor Processing Unit (TPU), a Neural Processing Unit (NPU), a Vision Processing Unit (VPU), a Machine Learning Accelerator, an Artificial Intelligence Accelerator, an Application Specific Integrated Circuit (ASIC), a Field Programmable Gate Array (FPGA), a Radio-Frequency Integrated Circuit (RFIC), a Neuromorphic Processor, a Quantum Processor, or any combination thereof.

(471) The processor **3102** may further be a multi-core processor having two or more independent processors (sometimes referred to as “cores”) that may execute instructions contemporaneously. Multi-core processors contain multiple computational cores on a single integrated circuit die, each of which can independently execute program instructions in parallel. Parallel processing on multi-core processors may be implemented via architectures like superscalar, VLIW, vector processing, or SIMD that allow each core to run separate instruction streams concurrently. The processor **3102** may be emulated in software, running on a physical processor, as a virtual processor or virtual circuit. The virtual processor may behave like an independent processor but is implemented in software rather than hardware.

(472) The mass storage device **3116** may include a machine-readable medium **3122** on which one or more sets of data structures or instructions **3124** (e.g., software) embodying or utilized by any of the techniques or functions described herein. The instructions **3124** may also reside, completely or at least partially, within the main memory **3104**, within the static memory **3106**, within the hardware processor **3102**, or the GPU **3103** during execution thereof by the machine **3100**. For example, one or any combination of the hardware processor **3102**, the GPU **3103**, the main memory **3104**, the static memory **3106**, or the mass storage device **3116** may constitute machine-readable media.

(473) While the machine-readable medium **3122** is illustrated as a single medium, the term “machine-readable medium” may include a single medium or multiple media (e.g., a centralized or distributed database and associated caches and servers) configured to store one or more instructions **3124**.

(474) The term “machine-readable medium” may include any medium that is capable of storing, encoding, or carrying instructions **3124** for execution by the machine **3100** and that causes the machine **3100** to perform any one or more of the techniques of the present disclosure or that is capable of storing, encoding, or carrying data structures used by or associated with such instructions **3124**. Non-limiting machine-readable medium examples may include solid-state memories and optical and magnetic media. For example, a massed machine-readable medium comprises a machine-readable medium **3122** with a plurality of particles having invariant (e.g., rest) mass. Accordingly, massed machine-readable media are not transitory propagating signals.

Specific examples of massed machine-readable media may include non-volatile memory, such as semiconductor memory devices (e.g., Electrically Programmable Read-Only Memory (EPROM), Electrically Erasable Programmable Read-Only Memory (EEPROM)) and flash memory devices; magnetic disks, such as internal hard disks and removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks.

(475) The instructions **3124** may be transmitted or received over a communications network **3126** using a transmission medium via the network interface device **3120**.

(476) Throughout this specification, plural instances may implement components, operations, or structures described as a single instance. Although individual operations of one or more methods are illustrated and described as separate operations, one or more of the individual operations may be performed concurrently, and nothing requires that the operations be performed in the order illustrated. Structures and functionality presented as separate components in example configurations may be implemented as a combined structure or component. Similarly, structures and functionality presented as a single component may be implemented separately. These and other variations, modifications, additions, and improvements fall within the scope of the subject matter herein.

(477) The embodiments illustrated herein are described in sufficient detail to enable those skilled in the art to practice the teachings disclosed. Other embodiments may be used and derived therefrom, such that structural and logical substitutions and changes may be made without departing from the scope of this disclosure. The Detailed Description, therefore, is not to be taken in a limiting sense, and the scope of various embodiments is defined only by the appended claims, along with the full range of equivalents to which such claims are entitled.

(478) Additionally, as used in this disclosure, phrases of the form “at least one of an A, a B, or a C,” “at least one of A, B, and C,” and the like should be interpreted to select at least one from the group that comprises “A, B, and C.” Unless explicitly stated otherwise in connection with a particular instance, in this disclosure, this manner of phrasing does not mean “at least one of A, at least one of B, and at least one of C.” As used in this disclosure, the example “at least one of an A, a B, or a C” would cover any of the following selections: {A}, {B}, {C}, {A, B}, {A, C}, {B, C}, and {A, B, C}.

(479) Moreover, plural instances may be provided for resources, operations, or structures described herein as a single instance. Additionally, boundaries between various resources, operations, modules, engines, and data stores are somewhat arbitrary, and particular operations are illustrated in the context of specific illustrative configurations. Other allocations of functionality are envisioned and may fall within the scope of various embodiments of the present disclosure. In general, structures and functionality are presented as separate resources in the example; configurations may be implemented as a combined structure or resource. Similarly, structures and functionality presented as a single resource may be implemented as separate resources. These and other variations, modifications, additions, and improvements fall within a scope of embodiments of the present disclosure as represented by the appended claims. Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense.

Claims

1. A computer-implemented method comprising: identifying audience parameter values, the audience parameter values indicating which users are members of an audience; determining attribute values for a plurality of attributes used to generate items based on the audience parameter values, the attribute values used to identify the members of the audience; generating a plurality of items for the audience based on the attribute values, wherein generating each item from the plurality of items comprises: creating a prompt based on a type of the item, the attribute values, and the audience parameter values, the prompt being a text description for generating new content;

selecting a generative artificial intelligence (GAI) tool to generate the item based on the type of item; and providing the created prompt to the selected GAI tool to obtain the item; causing presentation of the generated plurality of items in a user interface (UI); receiving on the UI a selection of one of the items from the generated plurality of items; and transmitting the selected item to one or more members of the audience.

2. The method as recited in claim 1, further comprising: providing a first user interface (UI) displaying information regarding different audiences, audience preferences, and content designed for each audience, the first UI providing an option to select and filter information by audience.

3. The method as recited in claim 2, wherein the first UI provides cards showing documents generated for each communication channel.

4. The method as recited in claim 1, further comprising: providing a second UI showing a performance of content presented to users and a feed of items available for future communications to the audience.

5. The method as recited in claim 4, wherein the performance is a measurement of a response of users to communications, and the performance is based on one or more of a number of views, a percentage of users that selected a link for additional information, a percentage of users that bought a product or service associated with a transmitted item.

6. The method as recited in claim 4, wherein generating each item further comprises utilizing past performance data to generate the item.

7. The method as recited in claim 1, wherein determining attribute values further comprises: utilizing an image-recognition model to determine features of images sent to users; determining a relevance score for the features based on tracked performance of the images sent to users; and selecting the attribute values based on the relevance score of the features.

8. The method as recited in claim 7, wherein the image-recognition model is trained using a dataset of images with known performance metrics, the image-recognition model learning correlations between visual elements and item performance, wherein the image-recognition model predicts a performance of new images based on visual features of the new images.

9. The method as recited in claim 1, further comprising: providing a second UI showing audience information on a feed, the second UI showing content organized by audience for a plurality of audiences, the second UI presenting the audience information in a table organized by calendar date, with each column for a different day and each row showing communications planned or delivered for each day for each audience.

10. The method as recited in claim 1, further comprising: generating a feed for the audience, wherein the items generated for the feed are deliverable via a plurality of channels that comprise publishing to social media, connecting to email automation systems, providing audience-level personalization on a webpage, and distributing through customer journey orchestrators.

11. A system comprising: a memory comprising instructions; and one or more computer processors, wherein the instructions, when executed by the one or more computer processors, cause the system to perform operations comprising: identifying audience parameter values, the audience parameter values indicating which users are members of an audience; determining attribute values for a plurality of attributes used to generate items based on the audience parameter values, the attribute values used to identify the members of the audience; generating a plurality of items for the audience based on the attribute values, wherein generating each item from the plurality of items comprises: creating a prompt based on a type of the item, the attribute values, and the audience parameter values, the prompt being a text description for generating new content; selecting a generative artificial intelligence (GAI) tool to generate the item based on the type of item; and providing the created prompt to the selected GAI tool to obtain the item; causing presentation of the generated plurality of items in a user interface (UI); receiving on the UI a selection of one of the items from the generated plurality of items; and transmitting the selected item to one or more members of the audience.

12. The system as recited in claim 11, wherein the instructions further cause the one or more computer processors to perform operations comprising: providing a first user interface (UI) displaying information regarding different audiences, audience preferences, and content designed for each audience, the first UI providing an option to select and filter information by audience.
 13. The system as recited in claim 12, wherein the first UI provides cards showing documents generated for each communication channel.
 14. The system as recited in claim 11, wherein the instructions further cause the one or more computer processors to perform operations comprising: providing a second UI showing a performance of content presented to users and a feed of items available for future communications to the audience.
 15. The system as recited in claim 14, wherein the performance is a measurement of a response of users to communications, and the performance is based on one or more of a number of views, a percentage of users that selected a link for additional information, a percentage of users that bought a product or service associated with a transmitted item.
 16. A non-transitory machine-readable storage medium including instructions that, when executed by a machine, cause the machine to perform operations comprising: identifying audience parameter values, the audience parameter values indicating which users are members of an audience; determining attribute values for a plurality of attributes used to generate items based on the audience parameter values, the attribute values used to identify the members of the audience; generating a plurality of items for the audience based on the attribute values, wherein generating each item from the plurality of items comprises: creating a prompt based on a type of the item, the attribute values, and the audience parameter values, the prompt being a text description for generating new content; selecting a generative artificial intelligence (GAI) tool to generate the item based on the type of item; and providing the created prompt to the selected GAI tool to obtain the item; causing presentation of the generated plurality of items in a user interface (UI); receiving on the UI a selection of one of the items from the generated plurality of items; and transmitting the selected item to one or more members of the audience.
 17. The non-transitory machine-readable storage medium as recited in claim 16, wherein the machine further performs operations comprising: providing a first user interface (UI) displaying information regarding different audiences, audience preferences, and content designed for each audience, the first UI providing an option to select and filter information by audience.
 18. The non-transitory machine-readable storage medium as recited in claim 17, wherein the first UI provides cards showing documents generated for each communication channel.
 19. The non-transitory machine-readable storage medium as recited in claim 16, wherein the machine further performs operations comprising: providing a second UI showing a performance of content presented to users and a feed of items available for future communications to the audience.
 20. The non-transitory machine-readable storage medium as recited in claim 19, wherein the performance is a measurement of a response of users to communications, and the performance is based on one or more of a number of views, a percentage of users that selected a link for additional information, a percentage of users that bought a product or service associated with a transmitted item.
-